Undergraduate Qualifications

B.Tech in Computer Science (AI and ML)

As per guidelines of the National Higher Education Qualification Framework (NHEQF)



School of Computer Science

(Academic Year: 2025- 2026)

Program Overview

The B.Tech program in Computer Science and Engineering (CSE) with a specialization in Artificial Intelligence (AI) and Machine Learning (ML) is designed to equip students with cutting-edge skills and practical experience. Over four years, students will build a strong foundation in programming languages like Java, master Data Structures and Algorithms, and gain expertise in the MERN stack (MongoDB, Express.js, React, Node.js) for full- stack web development.

In the final years, the focus shifts to advanced AI and ML, preparing students to tackle real-world challenges. The program emphasizes practical learning through hands-on labs, workshops, internships, and live projects, ensuring graduates are industry-ready.

By the end of the course, students will be well-prepared to excel in careers across software development, AI, ML, and related fields, driving innovation and technological advancement.

Program Learning Outcomes

Program Specific Outcomes	
	A graduate of B.Tech in CS and (AI/ML) should be able to:
PSO 1	Demonstrate the acquisition of comprehensive knowledge and coherent understanding of computer science, AI, and ML in a broad multidisciplinary context, including their foundational concepts, programming languages like python, Java, linkages with related fields such as data science and software engineering, and current and emerging developments in these technologies.
PSO 2	Demonstrate the acquisition of practical, professional, and procedural knowledge required for carrying out advanced tasks in AI, ML, and full-stack development, including proficiency in the MERN stack (MongoDB, Express.js, React, Node.js), implementing intelligent systems, undertaking self-employment initiatives, and fostering entrepreneurship in technology-driven enterprises.
PSO 3	Demonstrate the acquisition of skills in specialized areas of AI, ML, Data Structures, and Algorithms (DSA) within a broad multidisciplinary context, encompassing a wide range of practical skills for developing and deploying efficient software solutions and AI models in both routine and complex, variable contexts.
PSO 4	Demonstrate the capacity to extrapolate from learned concepts in AI, ML, Java, and full-stack development, apply acquired competencies to real-life situations, and generate innovative solutions, rather than simply replicating curriculum content, to address new and unfamiliar technological challenges.
General Specific Outcomes	
	A graduate of B.Tech in CS (AI/ML) should be able to:
PO 5	Demonstrate the capability for complex problem-solving, applying advanced techniques in AI, ML, and software development to design innovative solutions.
PO 6	Demonstrate the capability for critical thinking by analyzing, evaluating, and synthesizing information to make informed decisions in complex computational contexts.
PO 7	Demonstrate the ability for creativity in developing new algorithms, software applications, and AI models that push the boundaries of current technology.
PO 8	Demonstrate the skills that enable them to communicate effectively, both in technical documentation and collaborative environments, with peers, stakeholders, and non-technical audiences.
PO 9	Demonstrate the capability for analytical reasoning/thinking in deconstructing complex problems into manageable components and systematically addressing them using data-driven approaches.
PO 10	Demonstrate the ability for coordinating and collaborating with others, working effectively in multidisciplinary teams to achieve common goals in projects and research.
PO 11	Demonstrate the capability for leadership readiness, taking initiative in projects, guiding teams, and making strategic decisions in AI/ML endeavors.
PO 12	Demonstrate 'learning how to learn' skills, showing the ability to adapt to new technologies and methodologies, and continually update their knowledge base in the rapidly evolving field of CSE and AI/ML.
PO 13	Demonstrate the capability for digital and technological skills, mastering various programming languages, tools, and platforms relevant to AI, ML, and software development.

PO 14	Demonstrate multicultural competence and an inclusive spirit by respecting diverse perspectives and working effectively in global, multicultural environments.
PO 15	Demonstrate the acquisition of knowledge and attitude required for value inculcation, upholding ethical standards and integrity in the development and deployment of AI and software solutions.
PO 16	Demonstrate the ability for autonomy, responsibility, and accountability in managing individual projects and contributing to team outcomes with professionalism and diligence.
PO 17	Demonstrate the acquisition of and ability to apply knowledge, skills, attitudes, and values required for environmental awareness and action, particularly in developing sustainable and energy-efficient technologies.
PO 18	Demonstrate the capability to participate in community-engaged services/activities, applying technical expertise to projects that promote societal well-being and address community challenges.
PO 19	Demonstrate the ability to identify with or understand the perspective, experiences, or points of view of others, recognizing the ethical implications of AI/ML technologies on diverse populations and making decisions that consider the broader impact on society.

Abbreviation and Definition

Abbreviation	**Definition**
MDP	Multidisciplinary
AEC	Ability Enhancement Courses
VAC	Value Added Courses
SEC	Skill Enhancement Courses
MC	Major (Core)
MD	Major (Discipline)
MIP	Major (Industry Practice)
VETI	Vet (Industry Immersion)
MI	Minor (Electives)

Semester Wise Structure and Curriculum

Semester-1							
	Course Title	**Category**	**L**	**P**	**Pr**	**Credits**	
	Fundamental of Computer Programming (FOCP)	MC-1	0	3	0	3	
	Frontend Development Basics	MC-2	0	3	0	3	
	Problem Solving using Programming	MC-3	0	3	0	3	
	Mathematics for AI/ML	MC-4	3	0	0	3	
	Employability Skills (Basics)	SEC-1	3	0	0	3	
	On The Job Training-1	MIP-1	0	0	6	6	
	Total		**6**	**9**	**6**	**21**	

Semester-2							
	Course Title	**Category**	**L**	**P**	**Pr**	**Credits**	
	Data Structures and Algorithms	MC-5	0	3	0	3	
	Advance Frontend Development	MC-6	0	3	0	3	
	Database Management Systems	MC-7	0	3	0	3	
	Professional Skills (Team Skills)	SEC-2	3	0	0	3	
	Fundamentals of Business	MDP-1	3	0	0	3	
	On The Job Training-2	MIP-2	0	0	6	6	
			6	**9**	**6**	**21**	

Semester-3							
	Course Title	**Category**	**L**	**P**	**Pr**	**Credits**	
	Advance Problem Solving using Programming	MC-8	0	3	0	3	
	Full Stack Web Development	MC-9	0	4	0	4	
	Fundamentals of Data Science	MC-10	0	3	0	3	
	Mathematics for AI/ML - II	MC-11	3	0	0	3	
	Essentials of Digital Marketing	MDP-2	3	0	0	3	
	On The Job Training-3	MIP-3	0	0	6	6	
	Total		**6**	**10**	**6**	**22**	

Semester-4							
	Course Title	**Category**	**L**	**P**	**Pr**	**Credits**	
	Optimization and Advanced Techniques in DSA	MC-12	0	3	0	3	
	Operating System	MC-13	0	3	0	3	
	Fundamentals of Machine Learning	MC-14	0	4	0	4	
	Science and World	MDP-3	3	0	0	3	
	Universal Human Values	VAC-1	3	0	0	3	
	On The Job Training-4	MIP-4	0	0	6	6	
			6	**10**	**6**	**22**	

Semester-5							
	Course Title	**Category**	**L**	**P**	**Pr**	**Credits**	
	Advance Machine Learning	MC-15	0	4	0	4	
	Fundamentals of System Design	MC-16	0	3	0	3	
	Data Communication and Computer Networks	MC-17	0	3	0	3	
	Entrepreneurship Orientation	SEC-3	3	0	0	3	
	On The Job Training-5	MIP-5	0	0	6	6	
	Total		**3**	**10**	**6**	**19**	

Semester-6							
	Course Title	**Category**	**L**	**P**	**Pr**	**Credits**	
	Cloud Computing	MC-18	0	3	0	3	
	Advance System Design	MC-19	0	3	0	3	
	Fundamentals of Deep Learning	MC-20	0	4	0	4	
	Communication Skills (English)	AEC-1	0	4	0	4	
	Professional Skills (Career Skills)	SEC-4	3	0	0	3	
	On The Job Training-6	MIP-6	0	0	6	6	
			3	**14**	**6**	**23**	

Semester-7							
	Course Title	**Category**	**L**	**P**	**Pr**	**Credits**	
	Natural Language Processing	MC-21	2	2	0	4	
	Advance Deep Learning	MC-22	2	2	0	4	
	Business Communication	AEC-2	4	0	0	4	
	On The Job Training-7	MIP-7	0	0	8	8	
	Total		**8**	**4**	**8**	**20**	

Semester-8							
	Course Title	**Category**	**L**	**P**	**Pr**	**Credits**	
	Project Dissertation	MC-23	0	12	0	12	
	On The Job Training-8	MIP-8	0	0	8	8	
			0	**12**	**8**	**20**	

Curriculum (Course-wise) –
Semester 1:

Course Code		**Course Name**	Fundamental of Computer Programming (FOCP)	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Demonstrate proficiency in programming concepts and apply foundational skills to understand and implement object-oriented programming principles.	3
CLO2	Understand programming fundamentals, including JavaScript syntax, data types, and control flow	2
CLO3	Apply JavaScript control flow mechanisms to develop and solve basic algorithms.	3
CLO4	Create and manipulate JavaScript objects, and understand object-oriented programming principles.	4
CLO5	Use JavaScript functions to write modular and reusable code.	3
CLO6	Utilize advanced JavaScript features like promises and template literals for asynchronous operations.	5

Module - 1: Introduction to C

History and Features of C Language, Structure of a C Program, Keywords, Identifiers, and Constants, Data Types and Variables, Input and Output Functions, Operators and Expressions, Control Structures (if, if-else, switch, loops), Functions and Modular Programming, Recursion, Arrays (1D and 2D), Strings and String Functions, Introduction to Structures, Pointers and Pointer Arithmetic, Dynamic Memory Allocation, File Handling in C, Introduction to Structured Programming, Overview of Object-Oriented Programming Concepts (Classes, Objects, Inheritance, Polymorphism, Encapsulation), and Comparison of Structured and Object-Oriented Paradigms.

Module 2: Fundamentals of JavaScript Programming

Overview of Programming Languages, Overview of JavaScript, Concept of Syntax, Introduction to VS Code, Installation of Node.js and Code Runner to run JS programs, JavaScript is a Dynamically Typed Language, Variables, Constants, Data Types, Operators, Control Flow: Introduction to the Concept of Conditionals: if, if else, if else if, nested if else , switch case, Introduction to the Concept of a Loop/Iterative Calculations: For Loop, While Loop, Do While Loop, Infinite Loop, For...in, For...of, Break and Continue.

Module-3: Objects and cloning

Objects in JavaScript, Creating and Accessing Attributes in an Object, Dynamic Nature of Objects, Factory and Constructor Functions to Create Objects, Object Oriented Style of Programming, Value vs. Reference Types, Cloning an Object, Getter and Setter Functions in an Object.

Module-4: Functions in JavaScript

Functions, Function Declaration and Function Expression, function parameters and arguments, Higher Order Functions, Hoisting, Rest Operator, Default Parameters, Template Literals

Module-5: Array and its methods

Array, Array Operations, Array Methods, Builtin Objects and Methods, Math functions,
String methods, Date Object, Exception Handling, Try and Catch, Error Objects, Scope and Closures, Local vs. Global Scope, Let vs. Var, Understanding "this" Keyword, promises in JavaScript.

Reference Books & Suggested Readings

1. Flanagan, David. *JavaScript: The Definitive Guide*. 7th ed., O'Reilly Media, 2020.
2. Resig, John, and Bear Bibeault. *Secrets of the JavaScript Ninja*. 2nd ed., Manning Publications, 2016.
3. Osmani, Addy. *Learning JavaScript Design Patterns*. O'Reilly Media, 2012.JavaScript: The Definitive Guide by David Flanagan
4. *JS: The Right Way*. Official Website. http://jstherightway.org/.
5. *JavaScript Garden*. Official Website. http://bonsaiden.github.io/JavaScript-Garden/.
6. *Object Playground*. Official Website. http://objectplayground.com/.
7. *The Modern JavaScript Tutorial*. https://javascript.info/

Course Code		**Course Name**	Frontend Development Basics	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Understand the basics of web development, including HTML, CSS, and JavaScript, and their roles in frontend vs. backend development.	2
CLO2	Apply CSS to style web pages using different methods, selectors, and properties, and debug CSS issues in a browser.	3
CLO3	Create and manage HTML elements, including lists, links, tables, forms, and multimedia, using semantic HTML for structured and accessible content.	4
CLO4	Implement CSS layouts and manipulate the DOM with JavaScript to manage content, handle events, and enhance web interactivity.	5

Module 1: Overview of Web Development

- Overview of Web Development, Front-End vs. Back-End, Use of HTML, CSS, and JavaScript, Web Development Tools, Frameworks and Libraries, Responsive Design, HTML File Structure, HTML File Structure Demo, Headings and Paragraphs, Headings and Paragraphs Demo, Introduce Chrome Inspector, HTML Text Elements , The Strong element, The Bring Attention To element, The Emphasis element, The Idiomatic Text element, Nesting elements, Superscript and Subscript text.

Module 2: Overview of CSS

- Overview of CSS, why 'Cascading' and its importance for Web Development?, Demo a sample CSS snippet and show CSS debugging at browser, Different ways of writing CSS: Inline, Internal, External, CSS selectors: Class, Id, CSS properties / Basic Styling: font & text, Colors, measurement of units, styling text, adding borders, Styling list, Styling buttons, adding icons, margin, padding.

Module 3: important HTML elements

- HTML Lists: Ordered Lists, Unordered Lists, Nesting Lists
- HTML Links: Anchor elements, Relative links, Linking to Page elements, Mailto and Telephone links, Image tag, Alt attribute HTML Tables, HTML table overview, Table, tr and td elements, Table header, Table Caption, Colspan and Rowspan
- HTML Forms: Overview of HTML forms and inputs, Input element, Form element, Button

element, Email, url and number input element, Input label, Name attribute, Required attribute, Textarea element, Checkbox, range(slider) input, Dropdown with select and options, Radio button groups, Other input types.

- HTML Semantic Elements: The Span element, What's the point of a div, Introducing Semantic markup, Exploring Semantic elements
- Multimedia: Audio and video tags

Reference Books & Suggested Readings

1. Powell. Thomas A., *JavaScript: The Complete Reference*
2. Lemay. Laura, rafe colburn, jennifer kyrnin, *Mastering HTML, CSS & JavaScript Web*, BPB Publication, 2016
3. Vishvajeet. Sisodia, *Basic of Web Design, HTML, CSS3*, Centrum Press, 2014.
4. Zak Ruvalcaba Anne Boehm, *Murach's HTML5 and CSS3*, 3rdEdition, Murachs/Shroff Publishers & Distributors Pvt Ltd, 2016.
5. *Learn HTML*. GeeksforGeeks. https://www.geeksforgeeks.org/learn-html/.
6. *CSS Tutorial*. W3Schools. https://www.w3schools.com/Css/.
7. *Coding Fantasy*. https://codingfantasy.com/.

Course Code		**Course Name**	Problem solving using Programming	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Understand the fundamentals of Java, including the environment setup, program structure, basic syntax, and data types.	2
CLO2	Apply Java concepts to handle user input, use control flow statements, and manage methods and arrays for problem-solving.	3
CLO3	Analyze and utilize Java data structures, such as arrays, ArrayLists, HashMaps, and HashSets, and handle exceptions effectively	4
CLO4	Implement object-oriented programming principles in Java, including classes, inheritance, method overloading/overriding, and analyze time and space complexity.	5

Module-1: Introduction to Java

- Introduction to Java, JDK, JRE, Java environment setup in VS Code, The structure of a basic Java program, Statements, Indentation, methods, 2 types of parentheses, Class, How to run Java program. System.out.print and System.out.println, escape sequence(e.g. \n \t), Main function, calling another function in Main function, Order of statements in a method, Comments in Java, Introduction to Variables, Variable and Data Types, String Data Type, Type casting.

Module-2: User Input, Methods and Control flow

- User input: Scanner class, System.out.format, Tokens, and nextLine(), Conditional Statement, Operators, and Loops, for each loop, Lexical Scope, Introduction to Arrays and array methods, Introduction to methods, arguments, and parameters, and return statement.

Module-3: Data Structures

- Garbage Collection, and Memory leaks, Strings - Ways to create a String Literal, Immutability of Strings, StringBuffer, StringBuilder, Stack Memory Allocation and Heap Memory Allocation, Arrays vs ArrayList, HashMap and HashSet, Exception Handling in Java,

Module-4: Basics of Object Oriented Programming (OOPs)

- Object Oriented Programming Overview, Classes, Objects, Public and Static members, Method Overloading and Overriding, Inheritance, Introduction to Time and Space Complexity, an

Overview of Algorithms, Factors affecting Time Complexity, Worst, Average, and Best Case analysis, Input and Auxiliary Space.

Reference Books & Suggested Readings

1. J.C. Mitchell (1996), Foundations for Programming Languages, MIT Press.
2. D.P. Friedman, M. Wand (2008), Essentials of Programming Languages, MIT Press, 3rd Edition.
3. R. SETHI (2006), Programming Languages: Concepts and Principles, Pearson, 2nd Edition
4. Java Documentation. Oracle. https://docs.oracle.com/javase/tutorial/.

Course Code		**Course Name**	Mathematics for AI/ML	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO2	Understand and explain the key mathematical concepts and theories that underpin AI/ML algorithms, including linear algebra, probability, and statistics.	2
CLO3	Apply mathematical models and methods to solve complex AI/ML problems, such as optimizing machine learning algorithms and analyzing data sets.	3
CLO4	Develop and implement mathematical models to simulate AI/ML processes, enhancing their problem-solving abilities in real-world scenarios.	4
CLO5	Analyze the performance of AI/ML algorithms using mathematical reasoning, identifying potential areas for improvement.	5

Module 1: Linear Algebra and Vector Calculus

- Vectors and Matrices: Operations, properties, and applications in AI/ML.
- Eigenvalues and Eigenvectors: Principal Component Analysis (PCA).
- Vector Spaces: Concepts of basis, dimension, and linear independence.
- Matrix Decomposition: Singular Value Decomposition (SVD) and its applications.
- Gradient and Hessian: Introduction to vector calculus and its role in optimization.

Module 2: Probability and Statistics

- Probability Theory: Basics of probability, conditional probability, and Bayes' theorem.
- Random Variables and Distributions: Discrete and continuous distributions, expectation, and variance.
- Statistical Inference: Hypothesis testing, confidence intervals, and p-values.
- Regression Analysis: Linear regression, correlation, and causation.
- Bayesian Statistics: Introduction to Bayesian methods in AI/ML.

Module 3: Optimization Techniques

- Convex Optimization: Convex sets, functions, and optimization problems.
- Gradient Descent: Optimization algorithms, learning rates, and convergence.
- Lagrange Multipliers: Constrained optimization techniques.

- Stochastic Optimization: Methods like Stochastic Gradient Descent (SGD).
- Optimization in Neural Networks: Backpropagation and cost function minimization.

Module 4: Discrete Mathematics and Graph Theory

- Set Theory: Basics of sets, functions, and relations.
- Combinatorics: Permutations, combinations, and their applications in algorithm analysis.
- Graph Theory: Graphs, trees, and networks; shortest path algorithms.
- Markov Chains: State transitions and applications in AI/ML.
- Discrete Probability: Probabilistic methods in discrete spaces.

References Books & Suggested Readings

1. Lay, David C., Steven R. Lay, and Judi J. McDonald. *Linear Algebra and Its Applications*. 5th ed., Pearson, 2016.
2. Strang, Gilbert. *Introduction to Linear Algebra*. 5th ed., Wellesley-Cambridge Press, 2016.
3. Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong. *Mathematics for Machine Learning*. Cambridge University Press, 2020.
4. Grimmett, Geoffrey R., and David R. Stirzaker. *Probability and Statistics*. 3rd ed., Oxford University Press, 2001.
5. Bertsekas, Dimitri P. *Nonlinear Programming*. 3rd ed., Athena Scientific, 2016.
6. *Khan Academy: Linear Algebra*. https://www.khanacademy.org/math/linear-algebra.
7. *MIT OpenCourseWare: Probability and Statistics*. https://ocw.mit.edu/courses/mathematics/18-05-introduction-to-probability-and-statistics-spring-2014/
8. *GeeksforGeeks: Graph Theory Tutorial*. https://www.geeksforgeeks.org/graph-theory-basics/

Course Code		**Course Name**	Employability Skills-Basics	**Course Category**	SEC	Skill Enhancement Course	L	P	Pr	C
							3	0	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcome:

CLO No	At The End Of The Course, The Learner Will Be Able To	Blooms Taxonomy (Bt) Level
CLO1	Demonstrate proficiency in various forms of communication to achieve desired outcomes in personal and professional interactions	4
CLO2	Demonstrate proficiency in workplace communication skills and Negotiation techniques for efficient and productive communication within the workplace	4
CLO3	Describe teamwork dynamics within the team environment	2
CLO4	Identify different types of customers and effectively respond to their needs and inquiries, fostering a customer-centric approach	2
CLO5	Demonstrate personal and professional competencies for personal growth, organizational success, and overall well-being	4
CLO6	Demonstrate critical thinking, creative thinking, and decision-making skills to respond effectively to a variety of situations	4
CLO7	Demonstrate an understanding of cultural and gender diversity in the workplace to foster a culture of inclusivity and collaboration.	4
CLO8	Prepare for job interviews, including readiness, conclusion strategies, and post-interview follow-up, to excel in job interviews.	4

Module 1: Communication Basics

- **Communication: Basics And Importance** – Definition, Importance, Improving Communication Skills For Better Personal And Professional Relationships
- **Speaking: Greetings And Introductions** - Conversations, Types Of Conversations, Words Approach
- **Writing: Understand Personal Experiences And Thoughts** – How To Write A Paragraph, Craft Approach
- **Non-Verbal Communication** – Definition, Importance, Improving Non-Verbal Communication
- **Active Listening** – Definition, Respect Approach
- **Negotiation: Understanding Perspectives** – Definition, Open Approach
- **Self-Presentation: Making A Great First Impression** – Appear Approach
- **Self-Presentation: Excelling At Interviews** – What Do Recruiters Look For?, Perfect Approach
- **Self-Presentation: Rocking The Group Discussion** – Definition Of Group Discussion, Pitch Approach
- **Selling: Communicate Effectively To Gain Acceptance** – Communication Methods To Gain Acceptance For Product, Service, Idea, Accept Approach

Module 2: Communication Workplace

- **Speaking: On The Telephone** – Everyday Communications, Ways To Communicate Politely And Effectively On The Telephone, Words Approach
- **Speaking: Making Requests** – Ways To Request Politely, The Difference Between Permission And Request, Words Approach
- **Writing: Write Effective Notes** – Definition Of Note-Taking, Effectively Writing Notes, Organizing Notes, Pilot Approach, Cornell Method Of Note-Taking
- **Writing: Write Effective Emails** – Importance, The Difference Between Written And Verbal Communication, Pilot Approach
- **Negotiation: Negotiation In Action Getting To Yes** – Types Of Outcomes, Gains Approach

Module 3: Teamwork

- **Work Effectively In A Team** – Importance Of Working In A Team, Cause Model
- **Collaborate To Achieve Team Goals** – Importance, Identifying Goals, Success Model
- **Build Effective Relationships With Stakeholders** – What Is A Stakeholder?, Types Of Stakeholders, Treat Technique For Managing Stakeholders
- **Conflict Management: Identify And Resolve Conflicts** – Reasons For Conflicts, Techniques To Resolve Conflicts, Calm Approach

Module 4: Customer Centricity

- **Types Of Customers** – Potential, Past, Current, Types Of Personalities (Ocean)
- **Responding Effectively To Customers** – Importance, Last Approach

Module 5: Attitudes and Behavioral Skills

- **Time And Task Management: Plan And Manage Tasks Within A Timeline** – Time Management Definition, Planning And Managing Tasks, Staying Organized, Plot Approach
- **Time And Task Management: Plan, Priorities, And Manage Tasks** – Managing Time, Organizing And Prioritizing Tasks, Train Model
- **Quality Consciousness: Introduction To Quality** – Defining Quality Consciousness, The Importance Of Establishing Standards
- **Quality Consciousness: Understand The Impact Of Errors** – Defining Errors, The Impact Of Errors, Avoiding Mistakes
- **Result Orientation: Introduction To Result Orientation** – Defining Result Orientation, The Importance Of A Result-Oriented Mindset, How To Succeed And Get Desired Results?
- **Result Orientation: Plan Tasks To Achieve Goals** – Becoming Result-Oriented, Result Orientation Process
- **Self-Development - Positive Attitude** – Defining Attitude, The Importance Of Positive Mindset, Glad Model

- **Self-Awareness: Know Yourself** – Defining Self-Awareness, Identifying Strengths, Interests And Areas Of Improvement, Using Strengths To Achieve Goals
- **Responding To Change: Understand And Adjust To Change** – Defining Change, Reacting To Change, Ways To Adapt To Change
- **Personal Health, Hygiene, And Grooming** – The Importance Of Grooming, Importance Of Personal Hygiene, Grooming Essentials, Effects Of Not Maintaining Appearance
- **Adopting Safety Practices: Health, Environment, And Safety Awareness** – Ways To Stay Fit And Healthy, Keeping Surroundings Clean, Keeping Yourself Safe, Pass Technique For Fire Extinguishers
- **Gain Financial Literacy** – Importance Of Financial Literacy, Financial Tools In Daily Life, Components Of Salary, Types Of Bank Accounts, Importance Of Insurance

Module 6: Problem Solving

- **Problem Solving: Introduction To Critical Thinking** – Importance Of Critical Thinking, Applications, Analyzing Problems, Link Between Ideas, Reason Model
- **Problem Solving: Introduction To Creative Thinking** – Defining Creative Thinking, Importance, Applications
- **Problem Solving: Introduction To Decision Making** – Defining Decision Making, Importance, Elements Of Decision Making
- **Decision Making: Respond Effectively To A Situation** – Decision-Making Process, Application In Different Scenarios, Decide Model

Module 7: Workplace Awareness

- **Cultural Fitment & Gender Diversity** – Stereotyping And Unconscious Bias, Diversity Issues And How To Resolve Them, Features Of A Diverse Workplace, Behavioral Elements Of A Diverse Workplace
- **Identify And Align With High-Growth Sectors** – Types Of Skill Sets, Steps Before Applying For A Job,
- **Organizational Structure And Values** – Organizational Structure And Hierarchy, Organizational Values, Work Environment And Culture
- **Searching And Applying For Relevant Job** – Platforms For Finding Job Vacancies, Creating A Resume, Cover Letter Formats, Job Application Requirements

Module 8: Success In Job Interviews

- **How To Prepare For Job Interview - Gaining Confidence** – Describing Strengths And Qualities, Do's And Don'ts For Identifying Strengths, Researching Roles And Responsibilities, Knowing Your Resume, The Importance Of Knowing About The Company, Asking Questions To The Interviewer
- **How To Prepare For Job Interview - Getting Ready** – List Of Documents For The Interview, Behavior At The Interview Venue, Grooming And Appearance For The Interview
- **How To Conduct Yourself At The Venue** – How To Make A Good First Impression, Tips For Conducting Yourself Well At The Venue
- **How To Answer Questions During The Interview** – Making A Good Impression, Tips To Answer Questions Effectively

- **How To Effectively Conclude The Interview** – Asking Relevant Questions To The Interviewer
- **How To Follow-Up After The Interview** – Following Up To Know The Application Status, Accepting The Job Offer, Handling Negative Results
- **Ace Your Job Interview** – Identify And Describe Your Strengths And Weaknesses, Updating Resume, Mock Interviews, Tricks To Do Well In An Interview

Reference Books & Suggested Readings

1. *Employability Skills Book 1*. National Institute of Open Schooling. https://nios.ac.in/media/documents/350_employability_skills/350_Book_01.pdf.
2. *Fundamentals of Employability*. Alison. https://alison.com/course/fundamentals-of-employability#google_vignette.
3. *Employability Skills Part 1*. Bharat Skills. https://www.bharatskills.gov.in/pdf/E_Books/CTS/ES/Hindi/ES_Part1_62hour_Hindi.pdf.
4. *Employability Skills Part 2*. Bharat Skills. https://www.bharatskills.gov.in/pdf/E_Books/CTS/ES/Hindi/ES_part2_58hour_Hindi.pdf.

Semester 2

Course Code		**Course Name**	Data Structures and Algorithms	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Design and implement greedy algorithms to solve optimization problems, such as finding the shortest path, minimizing costs, and maximizing profits, by understanding the greedy choice property and proving the correctness of their solutions.	3
CLO2	Solve complex problems by breaking them down into smaller, self-similar subproblems and implementing recursive solutions in Java, analyzing the time and space complexity of recursive algorithms, and identifying and handling potential issues like stack overflow.	3
CLO3	Implement stack data structures using arrays and linked lists, understand the Last-In, First-Out (LIFO) principle, and apply stacks to solve real-world problems such as expression evaluation, function call handling, and backtracking algorithms.	3
CLO4	Implement queue data structures using arrays and linked lists, understand the First-In, First-Out (FIFO) principle, and apply queues to solve real-world problems such as breadth-first search, job scheduling, and message passing.	3
CLO5	Explain the concept of trees as a hierarchical data structure, implement basic tree operations like insertion, deletion, and traversal (depth-first and breadth-first), and apply trees to represent hierarchical data such as file systems, organizational structures, and decision trees.	3

Module 1: Stacks and Queues

Introduction to Stack, Stack Array Implementation, Stack Linked List Implementation, Applications of Stack, Introduction to Queue, Basic Operations of Queue, Different Types of Queue, Applications of Queue, Implementation of Queue using Arrays and Linked List

Module 2: Recursion

Introduction to Recursion, Recursion vs Iteration, Finite and Infinite Recursion, What is Tail Recursion, Tail Recursion Optimization, When to Use Recursion, Applications of recursion,

Module 3: Greedy Algorithms

Introduction to Greedy Algorithm, Key characteristics (greedy choice property, optimal substructure), When to consider a greedy approach, Common pitfalls and limitations

Module 4: Tree

Introduction to Tree Data Structure, Basic Terminologies, Representation of Tree Data Structure, Types of Tree data structures, Basic Operations of Tree Data Structure, Properties of Tree Data Structure. Introduction to Binary Tree, Properties of Binary Tree, Types of Binary Tree, Applications, Advantages and Disadvantages of Binary Tree, Binary Tree (Array implementation), Complete Binary Tree, Perfect Binary Tree

Reference Books & Suggested Readings

1. Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. Introduction to Algorithms. 3rd ed. Cambridge, MA: MIT Press, 2009.
2. Karumanchi, Narasimha. Data Structures and Algorithms Made Easy. 5th ed. CareerMonk Publications, 2016
3. Sedgewick, Robert, and Kevin Wayne. Algorithms. 4th ed. Boston: Addison-Wesley, 2011.
4. Kleinberg, Jon, and Éva Tardos. Algorithm Design. Boston: Pearson, 2005.

Course Code		**Course Name**	Advance Frontend Development	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Frontend Development Basics	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Perform asynchronous operations using Promises, callbacks, and async/await, efficiently fetching and managing data from APIs.	3
CLO2	Evaluate the differences between class and function components, the use of JSX, static versus dynamic websites, and the role of the virtual DOM in optimizing rendering efficiency.	4
CLO3	Implement conditional rendering, prop drilling, higher-order components, and lifecycle methods for enhanced component functionality.	3
CLO4	Manage state and hooks, including useState and useEffect, for effective dynamic content updates and side effect handling.	4

Module-1: Asynchronous JavaScript in React

Asynchronous Programming Fundamentals, Callback functions and their limitations, The Promise object and its states (pending, fulfilled, rejected), Chaining Promises with then() and catch(),Using async/await for cleaner asynchronous code, Error handling with try...catch blocks.

Module-2: React Introduction and Components:

Introduction to React, Components, Class Component vs Function Component, JSX , static vs dynamic websites, Fragments, Virtual DOM

Module-3: Conditional Rendering and Props

Conditional Rendering, Props, PropTypes, Prop Drilling, React Lists and Keys, Components LifeCycle method, Higher Order Component

Module-4: State Management with React Hooks

Introduction to State Management and Hooks, useState Hook, useState with previous state, useState with Array, useState with Object, UseEffect Hook, UseEffect after render, Dynamic Content Handling

Reference Books & Suggested Readings

1. Banks, Eve Porcello, and Alex Banks. *Learning React: Modern Patterns for Developing React Apps*. 2nd ed., O'Reilly Media, 2020.
2. Allen, K. Scott. *React in Action*. Manning Publications, 2018.
3. *React Official Documentation*. React. https://react.dev/.
4. *Material-UI Documentation*. MUI. https://mui.com/.
5. *Axios Guide*. Axios. https://axios-http.com/.

Course Code		**Course Name**	Database Management Systems	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Analyze DBMS architecture, the ER model, and relational database design principles to create effective database solutions.	4
CLO2	Apply relational algebra concepts, perform advanced SQL operations such as joins, subqueries, aggregation functions, and filtering to manage and query relational databases.	3
CLO3	Illustrate schema refinement by applying functional dependencies, Boyce-Codd Normal Form, and Third Normal Form to normalize databases into BCNF and 3NF.	2
CLO4	ACID properties, transaction states, Subqueries and indexing	2

Module-1: DBMS Architecture and Data Models

DBMS Introduction, DBMS Architecture (1, 2 and 3 Tier),File System v/s DBMS, ER Model, Enhanced ER Model, Minimization of ER Diagram, Relational Model, Keys (Candidate, Super, Primary, Alternate and Foreign), Anomalies in Relational Model, Mapping from ER Model to Relational Model, Strategies for Schema design, Schema Integration

Module-2: Relational Algebra and SQL Queries

Relationship Algebra Selection and Projection, Set Operations, Renaming, Introduction to SQL, Data Types, Basic SQL Commands, Joins, Subqueries, Aggregation Functions: SUM(), AVG(), COUNT(), MIN(), MAX(), Group by, having

Module-3: Schema refinement and Normal forms

Introductions to Schema Refinement, Functional Dependencies, Boyce-Codd Normal Forms, Third Normal Form, Normalization-Decomposition into BCNF Decomposition into 3-NF.

Module-4: Transactions and Indexing

Introduction to Transaction State, Transaction properties, ACID Properties, Subqueries, Indexing , Types of Indexes: Primary, Unique, Composite.

Reference Books & Suggested Readings:

1. "Database System Concepts" by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan
2. “Fundamentals of Database Systems" by Ramez Elmasri and Shamkant B. Navathe
3. "Database Internals" by Alex Petrov
4. "A Guide to Relational Database Design" by C. J. Date
5. "The ANSI/SPARC Framework for Database Management System Evolution" by ANSI/X3/SPARC Study Group on Data Base Management Systems
6. "A Performance Evaluation of Database Systems" by Jim Gray
7. ACM Digital Library
8. IEEE Xplore
9. VLDB Journal

Course Code		**Course Name**	Professional Skills (Team Skills)	**Course Category**	SEC	Skill Enhancement Course	L	P	Pr	C
							3	0	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcome:

CLO No	At the end of the course, the learner will be able to	Blooms Taxonomy (Bt) Level
CLO1	Explain the importance of team skills and list the key team skills	2
CLO2	Apply cognitive skills such as critical thinking, problem-solving and the ability to learn, for smooth and efficient functioning in the workplace	3
CLO3	Apply non-cognitive skills such as empathy, creativity, teamwork, collaboration, interpersonal skills, and resilience for smooth and efficient functioning in the workplace	3
CLO4	Use trust and collaboration while working in a team	3
CLO5	Display effective communication as team leaders and members for the proper functioning of the team	3

Module 1: Introduction To Team Skills

Overview And Importance – What Is A Team?, Characteristics Of Effective Teams, Team Skills, Why Team Skills Matter?, Benefits Of Effective Teamwork

Key Team Skills – Communication, Collaboration, Leadership, Problem-Solving, Time Management, Emotional Intelligence

Module 2: Cognitive Skills

Introduction To Cognitive Skills – Overview, Developing Cognitive Skills

Critical Thinking Skills – Characteristics Of Critical Thinkers, Critical Thinking At The Workplace

Problem Solving Skills – Stages Of Problem-Solving, Essential Skills For Problem-Solving, Problem-Solving Strategies, Obstacles In Problem-Solving

Ability To Learn - Why Learning Matters?, Strategies For Growth, Traits Of Successful LearnersModule 3: Non-Cognitive Skills

Module 3: Introduction To Non-Cognitive Skills - Meaning And Types

Empathy – Meaning, Stages Of Empathy, Strategies To Develop Empathy At Work

Teamwork - Meaning, Advantages, Factors Contributing To The Success Of The Team, Strategies To Deal With Conflict Among Team Members

Creativity – Meaning, Strategies To Increase Creativity, Creative Skills At The Workplace

Collaboration – Meaning, Types, Importance, Collaborative Learning

Resilience, Perseverance And Self Control – Meaning, Strategies For Building Resilience

Interpersonal And Social Skills – Variants Of Interpersonal Skills, Developing Interpersonal Skills

Module 4: Trust And Collaboration

Importance Of Trust In Creating A Collaborative Team – Definition, Importance
Strategies To Build Trust With Employees – Types Of Strategies

Stages Of Team Development – Forming, Storming, Norming, Performing, Breaking

Collaboration Vs Cooperation Vs Coordination Vs Communication – Definitions, Comparison, Examples
Criteria For Evaluation Of Trust And Collaboration In Teams - Admit Mistakes, Offering Trust, Active Listening, Affirmation, Decision Making, Visibility, Coaching And Monitoring, Care And Concern, Confidentiality
Agree To Disagree And Disagree To Agree – Spirit Of Teamwork – Meaning, Consensus Decision
Fear Of Being Judged And Strategies To Overcome Fear - Understanding The Fear Of Being Judged, Signs And Symptoms Of Social Anxiety Disorder, Strategies To Overcome Fear Or Social Anxiety

Module 5: Presentation Skills

Presentation: Meaning And Types – Meaning, Types, Presentation For Internal And External Communication
Presentation Strategies – Knowing The Purpose, Knowing The Audience, Opening And CLOsing
Ways To Improve Presentation Skills Over Time - Presenting And Public Speaking Tips

Module 6: Listening As A Team Skill

Introduction To Listening As A Team Skill - Listening Skill, Advantages Of Effective Listening Skills, Types Of Listening
Listening As A Team Member And Team Leader - Listening As A Team Leader, Listening As A Team Member, Improving Listening Skills
Uses Of Active Listening Strategies To Encourage Sharing Of Ideas - The Importance Of Active Listening In The Workplace, Strategies For Improving Active Listening Skills To Encourage Sharing Of Ideas

Module 7: Brainstorming

Brainstorming: The Meaning And Process - Meaning And Process, Procedure For Conducting Brainstorming, Importance Of Using The Brainstorming Technique, Types Of Brainstorming
Principles Of Documentation Of Team Session Outcomes - Harvesting Ideas, Identifying The Workable Gem, Transforming Ideas Into Action, Documenting The Outcome, Review And Reflection

Module 8: Social And Cultural Etiquettes

Etiquette – Meaning, Need For Effective Interpersonal Relationships
Aspects Of Social And Cultural/Corporate Etiquette In Promoting Teamwork - Social Etiquette, Cultural Etiquette And Its Role In Promoting Teamwork, Corporate/Professional Etiquette

Module 9: Internal Communication

Internal Communication: Meaning And Need - Meaning, Need For Internal Communication
Various Channels Of Transmitting Information To Team Members – Online, Offline

Reference Books & Suggested Readings

1. Learning Team Skills – Student Edition, by Ph.D. Bell, Arthur H. (Author), Ph.D. Smith, Dayle M.
2. Hayes, J. (2002). Interpersonal Skills at Work (2nd ed.). Routledge
3. Teamwork Skills: Communicating Effectively in Groups https://www.coursera.org/learn/teamwork-skills-effective-communication
4. People and Soft Skills for Professional and Personal Success Specialization https://www.coursera.org/specializations/people-and-soft-skills-for-professional-success.

Course Code		**Course Name**	Fundamentals of Business	**Course Category**	MDP	Multidisciplinary	L	P	Pr	C
							3	0	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcome:

CLO No	At the end of the course, the learner will be able to	Blooms Taxonomy (Bt)Level
CLO1	Describe various business forms, their advantages, legal requirements, and the registration process for establishing a business	2
CLO2	Describe the nature and significance of business economics and its role in informed decision-making	2
CLO3	Explain the principles and necessity of business ethics and the concept and importance of social responsibility	2
CLO4	Describe the concepts and processes of management, planning, organizing, directing, controlling, and the traits and styles of effective leadership	2
CLO5	Identify various organizational structures and explain their advantages	1

Module 1: Introduction to Business – 10 hrs

Different Types of Business Forms: What is Business?, Types of Business

Advantages of different business forms and Legal requirements for establishing different businesses: Advantages, Legal requirements

Registration process of a business - Steps involved, Necessary documents, Importance, Costs associated, Government agencies, Benefits, Common mistakes to avoid

Module 2: Economics and Business – 6 hrs

Nature and Significance: Meaning of business economics, Nature

Role of business economics in decision-making - Fundamentals, Cost-benefit analysis, Demand and supply analysis, Pricing strategies, Economic forecasting, External factors affecting business decisions

Module 3: Ethics and Social Responsibility – 6 hrs

Need of Business Ethics: Meaning, Principles, Need

Concept of Social Responsibility Need: Meaning, Need

Module 4: Management and Leadership – 16 hrs

Concept of management: Process, Nature, Importance, Scope

Planning: Importance, Characteristics, Process, Types of Plans

Organising: Concept, Steps, Principles, Importance

Directing and Controlling: Concept, Principles, Elements and Importance,

Controlling and its Features, Importance and Process of Controlling

Leadership: Concept, Meaning, Effective Leadership Traits, Leadership Styles

Module 5: Organizational Structure – 6 hrs

Types and advantages of different types of organizational structure: Organisational Structure, Types
Choosing the right type of organizational structure: Necessity, Right structure and Growth, Stages in Organisational Development, Organisational
Developmental Strategy, Organisational Practices in 5 phases of growth, Organisational success

Reference Books & Suggested Readings:

1. Management by James A. F. Stoner, R. Edward Freeman, Amitabh Deo Kodwani, Daniel R. Gilbert, Ranjeet Nambudiri
2. Fundamentals of Business https://pressbooks.lib.vt.edu/fundamentalsofbusiness4e/front-matter/using-this-book/

Semester 3

Course Code		**Course Name**	Advanced Problem Solving using Programming	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Problem Solving using Programming	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Accurately define a Binary Search Tree (BST) and its key properties (e.g., left subtree < node < right subtree).	3
CLO2	Analyze the time and space complexity of BST operations in different scenarios (best, average, worst case).	3
CLO3	Represent graphs using different data structures (e.g., adjacency matrix, adjacency list) and understand their trade-offs in terms of space and time complexity.	3
CLO4	Understand the fundamentals of binary representation and bitwise operators (AND, OR, XOR, NOT, shift operators)..	3
CLO5	Implement heap operations (insertion, deletion, heapify) efficiently using arrays or binary trees.	3

Module 1: Binary search tree

Binary Search Trees (BSTs):Definition, operations (insertion, deletion, search), applications (e.g., efficient searching, sorting), Binary Search Tree Variations: AVL Trees, Red-Black Trees (brief introduction), Binary Search Tree Traversal : Depth First Search and Breadth First Search.

Module 2: Bit Manipulation

Bitwise Operators: AND, OR, XOR, NOT, Shift Operators, Applications of Bit Manipulation: Efficient set operations, counting set bits, swapping variables

Module 3: Graph Representations & Traversals

Graph Definitions:Directed and undirected graphs, weighted and unweighted graphs, Graph Representations:Adjacency Matrix, Adjacency List, Graph Traversals:Depth-First Search (DFS), Breadth-First Search (BFS)

Module 4: Graph Algorithms

Shortest Path Algorithms:Dijkstra's Algorithm, Bellman-Ford Algorithm, Minimum Spanning

Tree Algorithms:Kruskal's Algorithm, Prim's Algorithm, Topological Sort

Module 5: Heap

Introduction to Heaps, Definition of a Heap:Min-heap, Max-heap, Heap-order property, Heap Representation, Array-based implementation of heaps, Heap Operations : Insertion (push), deletion (pop), peek (find minimum/maximum), Priority Queues, Heapsort

Reference Books & Suggested Readings

1. "Introduction to Algorithms" by Cormen, Leiserson, Rivest, and Stein
2. "Algorithms" by Robert Sedgewick and Kevin Wayne
3. Binary Search Trees (BSTs): https://www.geeksforgeeks.org/binary-search-tree-data-structure/[1]
4. Bit Manipulation:
   - GeeksforGeeks: https://www.geeksforgeeks.org/cpp-bitwise-operators/
   - "Hacker's Delight" by Henry S. Warren: A specialized book for in-depth bit manipulation techniques.
5. Graphs:
   - GeeksforGeeks: https://www.geeksforgeeks.org/graph-data-structure-and-algorithms/[2] (for definitions, representations, and traversals)
   - GeeksforGeeks: https://www.geeksforgeeks.org/problems/implementing-dijkstra-set-1-adjacency-matrix/1, https://www.geeksforgeeks.org/problems/distance-from-the-source-bellman-ford-algorithm/1, https://www.geeksforgeeks.org/kruskals-minimum-spanning-tree-algorithm-greedy-algo-2/,[3] https://www.geeksforgeeks.org/prims-minimum-spanning-tree-mst-greedy-algo-5/, https://www.geeksforgeeks.org/topological-sorting/[4] (for specific graph algorithms)
6. Heaps:
   - GeeksforGeeks: https://www.geeksforgeeks.org/heap-data-structure/ (for heap definitions, operations, and applications)

Course Code		**Course Name**	Full Stack Web Development	**Course Category**	MC	Major	L	P	Pr	C
							o	3	o	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Design, develop, and deploy a complete web application using Node.js (with Express.js), React.js, and MongoDB	3
CLO2	Implement efficient state management solutions using Redux or Recoil to handle complex application logic and data flow.	4
CLO3	Implement security best practices, including input validation, authentication (JWT), authorization, and data encryption, across the entire application stack.	3
CLO4	Utilize data fetching libraries like React Query to optimize data fetching and caching for improved performance.	4

Module 1: Deep Dive into Node.js and Express

NodeJS Architecture, V8 , Libuv internals , Advanced Node.js Concepts: Asynchronous programming with Promises and async/await , Streams and file system operations, Node.js modules and the NPM ecosystem Mastering Express.js: Middleware functions (request logging, authentication, error handling), Building RESTful APIs with Express.js, Handling HTTP methods (GET, POST, PUT, DELETE)

Module 2: MongoDB Mastery

MongoDB Fundamentals: Introduction to NoSQL databases and MongoDB, Basic data types: documents, arrays, embedded documents, CRUD operations, MongoDB with Node.js : Installing and connecting to a MongoDB database using Node.js driver, Working with Mongoose: Creating Mongoose schemas, Defining models, Performing CRUD operations with Mongoose

Module 3: React.js: Advanced Concepts

React Router:Nested routes, dynamic routing, route parameters, Route guards, navigation, and URL management, State Management: Redux: Store, actions, reducers, middleware , Recoil: Atoms, selectors, and state management with a more modern approach, Context API: Sharing data across components

Module 4: Full-Stack Development and Integration

Building a Full-Stack Application: Integrating React frontend with Express.js backend, Implementing user authentication and authorization (JWT), Handling user sessions and data persistence, Testing and Debugging:Unit testing with Jest and Enzyme, Integration testing for frontend and backend

Reference Books & Suggested Readings

1. MongoDB: The Definitive Guide" by Kristina Chodorow
2. Node.js Design Patterns" by Addy Osmani
3. Learning React" by Alex Banks & Eve Porcello
4. Fullstack React" by GraphQL
5. Designing Data-Intensive Applications" by Martin Kleppmann

Course Code		**Course Name**	Fundamentals of Data Science	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Explain key concepts like supervised/unsupervised learning, classification/regression, neural networks, and deep learning architectures.	3
CLO2	Demonstrate proficiency in using NumPy for numerical computations and Pandas for data manipulation and analysis.	4
CLO3	Identify patterns, trends, and anomalies in datasets using statistical methods and visualization techniques.	3
CLO4	Write clean, efficient Python code for data manipulation, analysis, and visualization.	4

Module 1: Foundations of AI/ML/DL and Python Programming

What is AI? Types of AI (Narrow, General, Super), Applications of AI in everyday life, Introduction to Machine Learning (ML): Supervised learning (classification, regression), Unsupervised learning (clustering, dimensionality reduction), Reinforcement learnin, Introduction to Deep Learning (DL):Neural networks, deep neural networks, Convolutional Neural Networks (CNNs)

Module 2:Python Programming

Data types, variables, operators, Control flow (if/else, loops), Functions, modules, and packages, Introduction to NumPy: Arrays, matrices, and vectorized operations, Array indexing, slicing, and reshaping, Linear algebra operations

Module 3: Data Manipulation and Visualization with Pandas and Matplotlib/Seaborn

Introduction to Pandas: Series and DataFrames, Data loading and reading (CSV, Excel), Data selection, filtering, and manipulation, Data aggregation and grouping, Data Visualization with Matplotlib : Basic plotting (line plots, scatter plots, bar charts), Customizing plots (labels, colors, legends), Subplots and multiple plots, Data Visualization with Seaborn : Statistical graphics and visualizations, Creating more visually appealing and informative plots, Exploring relationships between variables

Module 4: Data Preprocessing and Exploratory Data Analysis (EDA)

Data Cleaning: Handling missing values (imputation techniques), Dealing with outliers, Data type conversion and formatting, Data Transformation: Feature scaling (standardization,

normalization), Encoding categorical variables (one-hot encoding, label encoding).

Reference Books & Suggested Readings

1. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow by Aurélien Géron
2. Python Crash Course by Eric Matthes
3. Python for Data Analysis by Wes McKinney
4. Data Science Handbook by Jake VanderPlas
5. Introduction to Statistical Learning by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani

Course Code		**Course Name**	Mathematics for AI/ML - II	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Mathematics for AI/ML	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Apply core concepts of linear algebra, including vectors, matrices, eigenvalues/eigenvectors, and linear transformations, to solve problems in data science.	3
CLO2	Conduct descriptive statistical analysis to summarize and understand key characteristics of datasets.	4
CLO3	Utilize probability distributions (normal, binomial, etc.) to model and analyze data, make predictions, and understand uncertainty.	3
CLO4	Perform hypothesis testing (e.g., ANOVA, chi-square) to draw statistically significant conclusions from data.	4

Module 1: Linear Algebra Foundations

Systems of Linear Equations, Vectors and Vector Spaces, Matrices, Matrix Operations, Determinants, Eigenvalues and Eigenvectors, Linear Transformations, Vector Spaces and Inner Product Spaces, Matrix Decompositions, Linear Programming

Module 2: Probability and Statistics

Descriptive Statistics: Measures of central tendency (mean, median, mode), Measures of dispersion (variance, standard deviation, quartiles), Data visualization (histograms, box plots, scatter plots), Discrete distributions (Bernoulli, binomial, Poisson), Hypothesis Testing: t-tests, ANOVA, chi-square tests

Module 3: Calculus and its Applications in Data Science

Differential Calculus: Limits, derivatives, and their interpretations, Optimization problems (finding maxima and minima), Gradient descent algorithm, Integral Calculus: Definite and indefinite integrals, Applications in probability and statistics (e.g., finding areas under curves)

Module 4: Dimensionality Reduction and Advanced Optimization

Dimensionality Reduction Techniques: Principal Component Analysis (PCA), t-SNE (t-Distributed Stochastic Neighbor Embedding) , Advanced Optimization Algorithms: Stochastic Gradient Descent (SGD) and its variations (mini-batch SGD), Adam optimizer, Hyperparameter tuning and model selection

Reference Books & Suggested Readings

1. **Linear Algebra and Its Applications** by David C. Lay
2. **Introduction to Probability** by Dimitri P. Bertsekas and John N. Tsitsiklis
3. **Introduction to Statistical Learning** by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani
4. **Deep Learning** by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
5. **Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow** by Aurélien Géron

Course Code		**Course Name**	Essentials of Digital Marketing	**Course Category**	MDP	Multidisciplinary	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Explain the concepts of E-business and digital marketing	2
CLO2	Utilize digital market research as a management tool	3
CLO3	Create engaging digital content using storytelling techniques and a structured content creation framework	5
CLO4	Plan and execute SEO strategies to enhance content visibility	4
CLO5	Analyze consumer behavior, customer segmentation, and product advertising decisions using marketing analytics tools	4

Module 1: Introduction To Digital Marketing

E-Business, Digital Marketing, and International Business - Definition and differences between E-business and Digital Marketing, Global business models (B2B, B2C, C2C) and the role of digital platforms, The 4 Ps of marketing (Product, Price, Place, Promotion) in the digital world, Digital strategies for international business growth

Evolution and Importance of E-Business and Digital Marketing - Historical development and technological impacts on digital marketing, Digital transformation's effect on businesses and consumer behavior, The significance of digital marketing in scaling businesses globally, Competitive advantages offered by E-business and digital marketing

Marketing and Digital Economy - The relationship between digital marketing and global economic trends, Digital economy's influence on consumer behavior and business operations, Emerging trends in digital marketing and their economic impact

Module 2: Digital Market Research

Market Research in E-Business and Digital Marketing - Importance of market research for decision-making in digital strategies, Role of digital market research in understanding customer behavior, Tools and techniques for digital market research (surveys, web analytics, social media listening)

Data-Driven Market Research - Data collection methods and their relevance in digital research, Big data's role in analyzing market trends and consumer preferences, Ethical considerations in data usage for digital marketing

Emerging Trends in Digital Market Research - Use of AI and machine learning in predictive market research, Evolution of market research methodologies in the digital age, Integration of real-time data for dynamic marketing strategies

Module 3: Content marketing

Content Creation Framework and Storytelling - Developing a structured content creation strategy based on storytelling, Types of digital content (blogs, videos, infographics, social media posts), Importance of storytelling in engaging audiences and driving brand awareness

Strategizing for Content Marketing - Planning and generating content ideas that align with audience needs, Optimizing content for different digital platforms (websites, social media, video platforms), Creating high-impact blogs, videos, and user-generated content

Measuring Content Effectiveness - Key performance indicators (KPIs) for evaluating content success, Tools for measuring engagement (Google Analytics, SEMrush, social media analytics), Adjusting content strategies based on data insights and performance

Module 4: Search Engine Optimisation (SEO)

Introduction To SEO and its Importance - Basics of SEO and its relevance in improving content visibility, Key types of SEO (on-page, off-page, technical SEO), How search engines work: understanding search algorithms and indexing

SEO Strategies and Keyword Planning - Developing effective SEO strategies using keyword research, Balancing keyword optimization with content quality for readability, Integrating SEO with content marketing strategies

Advanced SEO Techniques - Local SEO strategies for location-based businesses, YouTube SEO: optimizing video content for higher rankings, Identifying and avoiding black hat SEO practices (keyword stuffing, link schemes)

Module 5: Marketing Analytics

Introduction to Marketing Analytics - Overview of digital marketing analytics and its role in strategy optimization, Key analytics tools (Google Analytics, SEMrush, CRM systems), Use of analytics in measuring the effectiveness of digital campaigns

Consumer Choice and Customer Analytics - Understanding consumer behavior through data and choice models, Customer segmentation and targeting using analytics insights, Personalizing marketing efforts based on customer behavior and preferences

Advertising and New Product Decisions - Analyzing the ROI of digital advertising (PPC, programmatic ads), Using A/B testing and analytics for product development and advertising strategies, Data-driven

Reference Books & Suggested Readings

1. Digital Marketing: Strategy, Implementation & Practice by Dave Chaffey and Fiona Ellis-Chadwick
2. Digital Business and Electronic Commerce: Strategy, Business Models and Technology by Bernd W. Wirtz
3. AI-Driven Marketing Research and Data Analytics edited by Richard S. Michalewicz
4. Predictive Analytics and Generative AI for Data-Driven Marketing Strategies by Hemachandran K, Debdutta Choudhury, Raul V. Rodriguez, and Jorge Wise
5. Epic Content Marketing by Joe Pulizzi
6. Building a StoryBrand: Clarify Your Message So Customers Will Listen by Donald Miller
7. The Art of SEO by Eric Enge, Stephan Spencer, and Jessie Stricchiola
8. SEO 2025: Learn Search Engine Optimization with Smart Internet Marketing Strategies by Adam Clarke
9. Marketing Strategy: Based on First Principles and Data Analytics by Robert W. Palmatier and Shrihari Sridhar
10. Data-Driven Marketing: The 15 Metrics Everyone in Marketing Should Know by Mark Jeffery

Semester 4

Course Code		**Course Name**	Optimization and Advanced Techniques in DSA	**Course Category**	MC	Major	L	P	Pr	C
							o	3	o	3
Pre-Requisite			Data Structures and Algorithms	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Define and explain the principles of Dynamic Programming: Overlapping subproblems and optimal substructure.	3
CLO2	Recognize when Dynamic Programming can be effectively applied to solve a problem.	4
CLO3	Analyze the time and space complexity of 1D 2DP solutions.	3
CLO4	Solve string-related problems using Dynamic Programming (e.g., string matching, substring problems).	4

Module 1: Introduction to Dynamic Programming

Principle of Optimality, Overlapping Subproblems, Memoization (Top-Down Approach), Tabulation (Bottom-Up Approach), Time and Space Complexity Analysis of DP solutions, Fibonacci Sequence, Coin Change Problem, House Robber Problem

Module 2: 1D and 2D Dynamic Programming

Longest Increasing Subsequence (LIS), Longest Common Subsequence (LCS), Minimum Cost Climbing Stairs, Edit Distance, 0/1 Knapsack Problem, Unbounded Knapsack Problem, Matrix Chain Multiplication

Module 3: Dynamic Programming on Trees and Strings

Tree Diameter, Maximum Path Sum in a Binary Tree, Tree DP for finding optimal solutions within a tree structure, Longest Palindromic Subsequence, String Matching Algorithms (e.g., Rabin-Karp), Word Break Problem

Module 4: Advanced Dynamic Programming and Problem Solving

Bitmasking Dynamic Programming, Digit DP, Convex Hull Trick, Practicing on LeetCode, HackerRank, and other competitive programming platforms

Reference Books & Suggested Readings:

1. Introduction to Algorithms by Cormen, Leiserson, Rivest, and Stein

2. Algorithms by Robert Sedgewick and Kevin Wayne
3. Grokking Algorithms by Aditya Bhargava
4. Competitive Programmer's Handbook by Antti Laaksonen
5. Elements of Programming Interviews in Java

Course Code		**Course Name**	Operating System	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Differentiate between different types of operating systems (batch, time-sharing, real-time, distributed) and their characteristics.	3
CLO2	Explain the role of the operating system as a resource manager (CPU, memory, I/O devices).	4
CLO3	Define and describe the process life cycle, including states (new, ready, running, waiting, terminated)	3
CLO4	Analyze and compare different process scheduling algorithms (e.g., FCFS, SJF, priority scheduling, round robin) and their impact on system performance.	4

Module 1: Introduction to Operating Systems

Definition and purpose of an operating system, Evolution of operating systems (batch, time-sharing, real-time, distributed), Role of the operating system as a resource manager (CPU, memory, I/O), Kernel and its functions, User interface (command-line, GUI), System calls and their importance

Module 2: Process Management

Definition of a process and its characteristics, Process states (new, ready, running, waiting, terminated), Process Control Block (PCB) and its fields, System calls for process creation and termination, Parent-child relationships, Scheduling criteria (CPU utilization, throughput, turnaround time, response time), Scheduling algorithms (FCFS, SJF, Priority Scheduling, Round Robin)

Module 3: Memory Management

Memory Management Techniques : Contiguous memory allocation (fixed partition, variable partition), Fragmentation (internal, external), Memory protection mechanisms, Paging and Segmentation: Paging: Concept of pages, page tables, translation lookaside buffer (TLB), Segmentation: Concept of segments, segment tables, Virtual memory and its implementation

Module 4: Deadlocks and I/O Management

Characteristics of a deadlock (mutual exclusion, hold and wait, no preemption, circular wait), Resource allocation graph, Deadlock prevention, avoidance, detection, and recovery techniques, I/O devices and drivers, I/O interrupts and interrupt handling, Buffering and caching techniques, Disk scheduling algorithms (FCFS, SSTF, SCAN)

Reference Books & Suggested Readings:

1. **Operating System Concepts by Abraham Silberschatz,** Peter B. Galvin, and Greg Gagne:
2. **Modern Operating Systems by Andrew S. Tanenbaum**
3. **Operating Systems**: Three Easy Pieces by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau.
4. **Linux Kernel Development by Robert Love**
5. **Operating System Design**: The Xinu Approach by Douglas Comer

Course Code		**Course Name**	Fundamentals of Machine Learning	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Differentiate between supervised, unsupervised, semi-supervised, and reinforcement learning.	3
CLO2	Understand the importance of data quality and the role of feature engineering in machine learning.	4
CLO3	Understand and implement techniques for model selection and hyperparameter tuning (e.g., cross-validation, grid search).	3
CLO4	Visualize data effectively using appropriate techniques to understand relationships between variables.	4

Module 1: Introduction to Machine Learning

What is Machine Learning , Types of Machine Learning: Supervised, Unsupervised, Semi-supervised, Reinforcement Learning, Key Concepts: Training, Testing, Validation, Generalization, Overfitting/Underfitting, Data in Machine Learning : Types of data (structured, unstructured), Data preprocessing (cleaning, transformation, feature engineering), Importance of data quality and bias

Module 2: Supervised Learning - Regression

Simple Linear Regression: Model equation, assumptions, and interpretation, Implementation in Python with scikit-learn, Evaluation metrics (R-squared, RMSE, MAE), Multiple Linear Regression: Multiple features, model interpretation, and feature selection, Implementation in Python with scikit-learn, Tree-Based Regression: Decision Trees: Algorithm, interpretation, and hyperparameter tuning, Random Forests: Ensemble learning, bias-variance trade-off, Implementation in Python with scikit-learn

Module 3: Model Selection and Evaluation

Model Selection Techniques: Train-test split, cross-validation (k-fold, stratified k-fold), Hyperparameter tuning (grid search, random search), Bias-variance trade-off, Model Evaluation Metrics : Regression metrics (R-squared, RMSE, MAE, Adjusted R-squared), Interpretation of evaluation metrics, Model comparison and selection

Module 4: Advanced Supervised Learning

Support Vector Machines (SVM) for Regression: SVM theory and applications in regression, Implementation in Python with scikit-learn, Ensemble Methods: Bagging, Boosting (e.g., Gradient Boosting Machines, XGBoost), Stacking, Implementation and evaluation of ensemble

methods

Reference Books & Suggested Readings:

1. **Machine Learning Specialization by Andrew Ng** https://shorturl.at/AmH1N
2. **Introduction to Statistical Learning by Gareth James**, Daniela Witten, Trevor Hastie, and Robert Tibshirani
3. **Elements of Statistical Learning by Trevor Hastie**, Robert Tibshirani, and Jerome Friedman:
4. **Machine Learning: A Probabilistic Perspective by Kevin P. Murphy**
5. **Python Machine Learning by Sebastian Raschka and Vahid Mirjalili**

Course Code		**Course Name**	Science and World	**Course Category**	MDP	Multidisciplinary	L	P	Pr	C
							3	0	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Discuss the philosophy and application of science and scientific thinking in daily life.	2
CLO2	Explain different concepts related to the ever-changing role of science and formulate coherent arguments	2
CLO3	Analyse the contribution of Modern science to the development of civilization and society	4
CLO4	Recognize the recent advancements and pressing challenges in the world due to Science and Technology	1
CLO5	Describe various domains of science across multiple cultures from a global perspective.	2

Module 1: Introduction to Science

Interpreting Science- Definition, Importance, Life without Science
Scientific Thinking- Definition, Steps of Scientific Methods, Concept of Controlled Experiments, Types of Variables i.e. Independent, Dependent, Controlled Variable, Steps of Controlled Experiments, Fact, Hypothesis, Theory, Law
Science and Technology- Difference between Science, Technology, Engineering and their interdependence, Technology Adoption Rates
Globalisation- Definition, Social, Economic and Technology Impact, Evolution of Computer Systems, Technology Driven Economy, Case Study: Covid-19 Economic Challenges

Module 2: Modern Science and Society

Theory of Evolution- Definition, Darwin's Theory of Evolution, Survival of the Fittest, Key to Evolution, Evolutionary History, Evidence of Evolution such as Fossil Records, Comparison of Anatomy, Biogeography, Molecular Biology
Genetics and Human Health- Fundamental of Gene, Traits, Heredity, and Genetics, Location of DNA in Humans, Mendel's Theory of Inheritance, Monohybrid Cross, Dihybrid Cross, Dominant Gene, Recessive Gene, Independent Assortment, Modern Applications of Genetics, Genetic Mutation, Causes of Mutation, Genetic Disorders and their types, Various genetic disorders like Hemophilia, Down Syndrome, Cancer, Pedigree Analysis.

Discovery of Antibiotics- Definition, Bacteria and its types, Disease-causing Bacteria, History of Antibiotics, Classes of Antibiotics, Mode of Action of Antibiotics against Bacteria, Application of Antibiotics, Difference between drugs and medicine, Antimicrobial Resistance (AMR), Prophylaxis, Misuse and overuse of antibiotics, Discovery Void, Prevention against AMR

Space Science- Definition, Outcomes of Space Science, Discoveries in Space Science, Need, Galaxy, Challenges of Space and Space Science, Evolution of Rocket Science, Historical Overview of Space Exploration, Various Government Space Agencies, Satellite Technology, Various types of Satellites, Need of Space Satellites, ISRO, Indians in Space Science, Types of Launch Vehicle for ISRO, Various Indian Space Missions like Chandrayaan 1, Chandrayaan 2, Mangalyaan, Gaganyaan, Space Junk, Space Tourism

Nanotechnology and Smart Materials- Nanotechnology Definition, Occurrence of Nanomaterials in Nature, Application of Nanotechnology, Smart Materials such as synthetic spider web, silk, and graphene, Challenges in Nanotechnology

Module 3: Science Technology and Life

India's Agriculture Development- Green Revolution in India 1960, Krishonnati Yojana, Mechanised Farm Tools, Irrigation System, Pesticide and fertilizers, High Yielding a variety of crops such as Sharbati variety, "New Leaf" variety, "IR8" variety, Environmental Aspects, Drawbacks such as Steps toward the Death Valley, Drying of Water bodies; Genetic property change in food grains

Information Revolution and Data Science- Evolution of Internet, Computer, Cyber-attacks such as Malware, Phishing, DDoS Attack, Man-in-the-middle Attack, Drive-by Download Attack

Industrial Revolution and Cyber-Physical World- History of Industrial Revolution, Artificial Intelligence, Types of AI, Applications of AI, Machine Learning, Types of ML, Applications of ML, Key Algorithms and techniques such as Linear regression, Decision tree, K-means clustering, Deep learning

Energy Issues and Renewable Energy Sources- Energy and its types, Advantages and Disadvantages of various forms of energy, Challenges, Hydrogen as a Fuel, Hydrogen Extraction from Different Sources, Application of hydrogen fuel, Working of Hydrogen Fuel, Challenges of hydrogen and fuel cell, Energy Usage in India and various countries

Environment, Sustainability and Survival- Climate Change, Causes of Environmental Change, Global Warming and its causes, Effect of Global Warming on Climate Change, Effect of Global Warming, Locust Attack in India, Glacial Lake Outburst Floods, Transboundary Pollution, Impacts on marine ecosystems, Solution for Climate Change, Environmental and climate change conferences

Biomimicry and Nature inspired design- Concept of Biomimicry, Application of Biomimicry in Technology like Bullet Trains, Biomimicry Design Process, Biomimicry Types

Reference Books & Suggested Readings:

1. B. Barnes and D. Edge (eds.), Science in Context: Readings in the Sociology of Science, The Open University Press, 1983.
2. G. Basalla, The Evolution of Technology, Cambridge University Press, 1988.
3. C.A. Alvares, Homo Faber: Technology and Culture in India, China and the West,

1500 to the Present Day, Allied Publishers, 1979.
4. A.F. Chalmers, What is this Thing called Science?, The Open University Press, 1980.

Course Code		**Course Name**	Universal Human Values	**Course Category**	VAC	Value Added Course	L	P	Pr	C
							3	0	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Explain the importance of living a harmonious life aligned with universal human values	2
CLO2	Discuss the vast potential of human beings and their responsibility to the universe on its account	2
CLO3	Develop universal human values and practice them consciously to be good human beings	3
CLO4	Conduct oneself in alignment with the universal human values while dealing with the ways of the world	3

Module 1: Introduction to Universal Human Values

Concept of Universal Human Values – Overview, What are values?, Human values, What are universal human values?
Relation with holistic living – What is holistic living?, Relation of universal human values and holistic living

Module 2: Living in Harmony

Living in harmony - as a human - What is Living in Harmony?, Life in harmony, What does living in harmony look like for an individual?
Living in harmony - as a family - Key roles of shared values, Shared values of families
Living in harmony - as a society and a race – Respect, Equality, Kindness, Honesty, Safety, Diversity, Cooperation, Environmental Care, Freedom

Module 3: Human Potential

Human potential – individual - How do we unlock human potential?, How to identify our potential?
Human potential – collective – Overview, Collaboration and working together
Impact of individual self on environment - Ripple Effect of Your Interactions
Impact of social group on their environment – Impact of family on environment, Impact of peer group on environment
Who is responsible? - Harmful impact of humans, Positive impact of humans

Module 4: Developing Universal Human Values

Introduction to Developing Universal Human Values - Developing Human Values, Self Reflection, Educate Yourself, Promote Open-Mindedness, Volunteer and Service, Emulate Role Models in Actions, Engage in Dialogue, Develop Global Perspective
Love and Compassion - Love and its forms, Love, Compassion and Inter-relatedness, The greatest proponents of Love and Compassion, Practicing Love and Compassion
Truth - Introduction to Truth, Great Individuals who are remembered for their value of truth, Practising Truth
Non-Violence - Introduction to non-violence, Important people who followed and propagated non-violence, Practising non-violence
Service - Introduction to service, Various forms of Service
Constitutional Values, Justice and Human Rights – Fundamental Values, Fundamental Rights, Fundamental Duties, Patriotism, Pride and Gratitude for the Nation
Good Practices - Self Worth, Self-Care, Holistic Living, Mindfulness and Meditation, Self-Reflection, Journal Writing, Impact Assessment

Module 5: Common Scenarios

Routine Scenarios – Love and Compassion based scenarios, Truth based scenarios, Non-violence based scenarios, Peace based scenarios, Service based scenarios, Renunciation or Sacrifice based scenarios
Life-changing Scenarios – Career Dilemma, Relationship Conflict, Health Crisis, Moral and Ethical Dilemma, Personal Loss, Financial Crisis

Reference Books & Suggested Readings:

1. Schwartz, Shalom H. **Value Orientations: Measurement, Antecedents, and Consequences Across Nations**. Springer.
2. Veugelers, Wiel. **Education and Human Values: Reconciling Talent with an Ethics of Care**. Sense Publishers.
3. Govindarajan, Vijay, and Chris Trimble. **The Other Side of Innovation: Solving the Execution Challenge**. Harvard Business Review Press.
4. Covey, Stephen R. **The 7 Habits of Highly Effective People: Powerful Lessons in Personal Change**. Simon & Schuster.
5. Harris, Sam. **The Moral Landscape: How Science Can Determine Human Values**. Free Press.

Semester 5

Course Code		**Course Name**	Advance Machine Learning	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Fundamentals of Machine Learning	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Implement and interpret Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Naive Bayes, Decision Trees, and Random Forest algorithms for classification tasks.	3
CLO2	Analyze the strengths and weaknesses of each classification algorithm and select the most appropriate algorithm for a given problem.	4
CLO3	Implement and interpret K-means and hierarchical clustering algorithms.	3
CLO4	Conduct thorough EDA to understand the characteristics of the data, identify patterns, and prepare data for model training.	4

Module 1: Supervised Learning: Classification

Classification Problem: Definition of classification, Key concepts: features, labels, training data, test data, Logistic Regression: Model equation, decision boundary, Implementation in Python with scikit-learn, Evaluation metrics (accuracy, precision, recall, F1-score, AUC-ROC), K-Nearest Neighbors (KNN): Distance metrics (Euclidean, Manhattan), Choosing the value of K, Implementation in Python with scikit-learn

Module 2: Advanced Classification Algorithms

Support Vector Machines (SVM): Linear SVM, kernel trick, Different kernel functions (linear, RBF, polynomial), implementation in Python with scikit-learn, Naive Bayes: Bayes' theorem and its application to classification, Naive Bayes assumption, Implementation in Python with scikit-learn, Decision Trees: Decision tree construction (ID3, C4.5), Overfitting and pruning, Implementation in Python with scikit-learn

Module 3: Ensemble Methods and Model Selection

Ensemble Methods: Random Forest: Bagging and feature randomness, Advantages of ensemble methods over single models, Implementation in Python with scikit-learn, Model Selection and Evaluation: Train-test split, cross-validation (k-fold, stratified k-fold), Hyperparameter tuning

(grid search, random search), Choosing the best model based on evaluation metrics, Handling Imbalanced Data, Techniques for dealing with imbalanced datasets (oversampling, undersampling, SMOTE)

Module 4: Unsupervised Learning: Clustering

K-means Clustering: Algorithm, choosing the number of clusters (elbow method), Implementation in Python with scikit-learn, Limitations of K-means, Hierarchical Clustering: Agglomerative and divisive clustering, Dendrograms and visualization, Implementation in Python with scikit-learn,

Reference Books & Suggested Readings:

1. **Hands-On Machine Learning with Scikit-Learn**, Keras & TensorFlow by Aurélien Géron
2. **Introduction to Statistical Learning by Gareth James**, Daniela Witten, Trevor Hastie, and Robert Tibshirani
3. **Pattern Recognition and Machine Learning** by Christopher M. Bishop
4. **Data Science from Scratc**h: First Principles with Python by Joel Grus
5. **Mining of Massive Datasets by Jure Leskov**ec, Anand Rajaraman, and Jeff Ullman

Course Code		**Course Name**	Fundamentals of System Design	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Apply core OOP concepts like encapsulation, abstraction, inheritance, and polymorphism in the design of software systems.	3
CLO2	Design and implement well-structured and maintainable classes and interfaces using OOP principles	4
CLO3	Identify appropriate design patterns (e.g., Singleton, Factory, Observer) to solve recurring design problems and improve code reusability, maintainability, and flexibility.	3
CLO4	Design relational database schemas using normalization techniques (1NF, 2NF, 3NF) to ensure data integrity and minimize redundancy.	4

Module 1: Object-Oriented Programming Fundamentals

Core OOP Concepts: Encapsulation,Abstraction,Inheritance,Polymorphism, Classes and Objects: Defining classes with attributes and methods, Creating objects and interacting with them, Access modifiers (public, private, protected),Interfaces and Abstract Classes: Defining interfaces and abstract classes,Implementing interfaces and abstract classes,Benefits of using interfaces and abstract classes

Module 2: Design Patterns

Creational Patterns:Singleton Pattern,Factory Pattern,Abstract Factory Pattern, Structural Patterns: Adapter Pattern, Decorator Pattern,Facade Pattern

Module 3: API Design and Integration

RESTful API Design Principles: HTTP methods (GET, POST, PUT, DELETE), Designing RESTful API endpoints and resources, API versioning and documentation, API Integration: Consuming APIs using libraries (e.g., Axios, Retrofit), Handling API requests and responses, Authentication and authorization in APIs, Error handling and rate limiting

Module 4: Database Schema Design

Relational Database Concepts: Tables, columns, rows, primary keys, foreign keys, Relationships between tables (one-to-one, one-to-many, many-to-many), Database Normalization : First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), Normalization process and its benefits, Designing normalized database schemas

Reference Books & Suggested Readings:

1. **Design Patterns:** Elements of Reusable Object-Oriented Software by Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides (commonly known as the "Gang of Four"[1] book
2. **Head First Design Patterns by Eric Freema**n, Elisabeth Robson, Bert Bates, and Kathy Sierra:
3. **Clean Code**: A Handbook of Agile Software Craftsmanship by Robert C. Martin
4. **Designing Data-Intensive Applications by Martin Kleppmann**
5. **RESTful Web Services by Leonard Richardson and Sam Ruby**

Course Code		**Course Name**	Data Communication and Computer Networks	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Understand and differentiate between various types of transmission media (e.g., wired, wireless, fiber optic).	3
CLO2	Explain fundamental networking terms (client, server, peer, host, bandwidth, packet, frame, protocol).	4
CLO3	Understand the characteristics of different network types (LAN, WAN, MAN).Relate real-world networking scenarios to the OSI model.	3
CLO4	Understand how data is encapsulated and decapsulated as it travels through the OSI layers.	4

Module 1: Introduction to Computer Networks

Fundamentals: What is a network , Basic terminology: Client, Server, Peer, Host, Node, Data communication concepts: Bits, bytes, bandwidth, latency, Network Types : LAN, WAN, MAN, Intranet, Extranet, Internet, Communication Models : Client-server model, Peer-to-peer model

Module 2: Network Components and Media

Network Devices: Hubs, switches, routers, modems, gateways, Functions and comparisons of these devices, Transmission Media, Guided media: Copper cables (twisted pair, coaxial), Fiber optic cables, Unguided media: Radio waves, microwaves, infrared, satellite, Characteristics of transmission media (bandwidth, speed, distance)

Module 3: Network Topologies and Communication

Network Topologies: Bus, star, ring, mesh, hybrid topologies, Advantages and disadvantages of each topology, Communication Types: Unicast, broadcast, multicast communication, Addressing schemes (MAC addresses, IP addresses)

Module 4: The OSI Model

Layers of the OSI Model: Physical Layer, Data Link Layer, Network Layer, Transport Layer, Session Layer, Presentation Layer, Application Layer, Functions of each layer, Data Encapsulation and Decapsulation: How data is encapsulated and decapsulated at each layer, Understanding headers and trailers

Module 5: Network Protocols and Data Flow

TCP/IP Protocol Suite: Overview of the TCP/IP model, TCP and UDP protocols, IP addressing (IPv4, IPv6), Packet Switching and Routing: How packets travel through a network, Role of

routers in packet forwarding, Basic routing concepts (routing tables, routing algorithms)

Reference Books:

1. **Computer Networking:** A Top-Down Approach by James F. Kurose and Keith W. Ross
2. **Data Communications and Networking by Behrouz A. Forouzan**
3. **TCP/IP Illustrated by W. Richard Stevens**
4. **Networking All-in-One For Dummies by Ed Tittel and Earl Morris**
5. **CCNA 200-301 Official Cert Guide Library** (For those, interested in pursuing Cisco certifications)

Course Code		**Course Name**	Entrepreneurship Orientation	**Course Category**	MDP	Multidisciplinary	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Explain the fundamental concepts of entrepreneurship	
CLO2	Identify different types of customers and effectively respond to their needs to enhance the overall customer experience	
CLO3	Analyze successful entrepreneurs' traits, and their risk-taking approaches, and cultivate leadership skills	

Module 1: Customer Centricity – 14 hrs

Introduction to Customer Centricity - Definition and Importance, Customer-Centric Business Models, Attitude and Approach, Continuous Learning and Adaptation

Types of Customers - Demographic Segmentation, Psychographic Segmentation, Behavioural Segmentation, Geographic Segmentation

Responding Effectively to Customers - Active Listening, Effective Communication, Conflict Resolution

Building Rapport with Customers - Trust Building Techniques, Personalization in Customer Service

Exceeding Customer Expectations - Anticipating Needs, Delivering More Than Promised

Enhancing Customer Experience - Customer Journey Mapping, Feedback and Improvement

Module 2: Entrepreneurial Mindset – 32 hrs

Recognize Opportunity - Opportunity Identification, Market Gaps and Trends

Creativity and Innovation - Idea Generation Techniques, Innovation Processes

Conception and Ideation - Concept Development, Idea Validation

Taking Risks - Risk Propensity, Managing Uncertainty

Self Confidence and Resilience - Building Self-Esteem, Overcoming Challenges

Success and Failure Stories of Famous Entrepreneurs

Never Give Up - Perseverance, Motivation Techniques

Getting Money For Business - Funding Options, Pitching to Investors

Dream and Achieve - Vision Setting, Goal Planning

Leadership and Team Spirit - Leadership Styles, Building Effective Teams

Serving the Society - Corporate Social Responsibility, Social Impact of Business

Taking Ownership - Accountability, Initiative and Responsibility

Adapt to Change - Flexibility, Change Management

Discover Yourself - Self-Assessment, Personal Growth Plans

Reference Books & Suggested Readings:

1. Entrepreneurship: Creating the Business https://alison.com/course/entrepreneurship-creating-the-business-revised
2. Essentials of Entrepreneurship: Thinking & Action https://www.coursera.org/learn/entrepreneurial-thinking
3. Effective Entrepreneurial Management: Strategy, Planning, RiskManagement, and Organization - Robert D. Hisrich • VelandRamadani, Springer (2017)
4. Entrepreneurship- Theory, Process Practice –by Kuratko &Hodgetts, Thompson South-Western Publication
5. Entrepreneurship –by Robert D. Hisrich (Edition-9)

Semester 6

Course Code		**Course Name**	Cloud Computing	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Define cloud computing and differentiate it from traditional IT.Explain the key characteristics of cloud computing (e.g., on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service).	3
CLO2	Analyze the advantages and disadvantages of each deployment model in different scenarios. Understand the concept of cloud bursting and its benefits.	4
CLO3	Identify and analyze use cases for different cloud services. Describe the key components of a cloud infrastructure (e.g., virtualization, networking, storage).	3
CLO4	Explain key security measures and best practices for cloud computing (e.g., access control, encryption, data loss prevention).	4

Module 1: Introduction to Cloud Computing

Fundamentals of Cloud Computing:Definition of cloud computing and its key characteristics, Benefits of cloud computing (cost-effectiveness, scalability, flexibility, etc.), Limitations of cloud computing, Cloud Computing Models: IaaS (Infrastructure as a Service): Virtual machines, storage, networking, PaaS (Platform as a Service): Development and deployment platforms, SaaS (Software as a Service): On-demand software applications

Module 2: Cloud Deployment Models and Architecture

Cloud Deployment Models: Public clouds,Private clouds, Hybrid clouds, Multi-cloud, Cloud Architecture: Core components of a cloud infrastructure (compute, storage, networking), Virtualization technologies, Load balancing, scaling, and high availability,

Module 3: Cloud Services

Compute Services: Virtual machines (VMs), containers (Docker, Kubernetes), Serverless computing (AWS Lambda, Google Cloud Functions), Storage Services: Object storage (Amazon S3, Google Cloud Storage), Block storage (EBS, Persistent Disks), File storage (EFS, Google Cloud Filestore), Database Services: Relational databases (RDS, Cloud SQL), NoSQL databases (MongoDB Atlas, Cloud Datastore), Data warehousing and analytics services

Module 4: Cloud Security and Compliance

Cloud Security Threats:Data breaches, DDoS attacks, malware, insider threat, Shared responsibility model in cloud computing, Cloud Security Measures: Access control (IAM, IAM roles), Encryption (data encryption at rest and in transit), Virtual Private Cloud (VPC), Compliance and Regulations:Industry-specific regulations (HIPAA, GDPR, PCI DSS), Compliance certifications (ISO 27001, SOC 2)

Module 5: Introduction to Major Cloud Providers

AWS (Amazon Web Services), Google Cloud Platform (GCP), Azure (Microsoft Azure):

Reference Books:

1. **Cloud Computing: Concepts, Technology & Architecture by George Reese**
2. **AWS Certified Solutions Architect Official Study Guide by John A. Rankin**
3. **Google Cloud Certified Professional Cloud Architect Study Guide by Google**
4. **Cloud Native Java:** Designing Resilient Systems with Spring Boot, Spring Cloud, and Cloud Foundry by Josh Long and Kenny Bastani
5. **Designing Data-Intensive Applications by Martin Kleppmann**

Course Code		**Course Name**	Advance System Design	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Fundamentals of System Design	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Design scalable architectures using techniques such as horizontal/vertical scaling, load balancing, caching, and distributed systems.	3
CLO2	Choose and implement appropriate load balancing algorithms (e.g., round robin, least connections, least response time).	4
CLO3	Evaluate the effectiveness of load balancing and caching strategies in improving system performance and scalability.	3
CLO4	Understand the benefits and challenges of microservices architecture (e.g., increased complexity, inter-service communication).	4

Module 1: Foundational Concept

Core Principles of System Design: Scalability: Horizontal and Vertical Scaling, Capacity Planning, Availability and Reliability: Fault Tolerance, Redundancy, High Availability, Performance: Latency, Throughput, Responsiveness, Maintainability: Modularity, Testability, Observability, Architectural Styles: Monolithic vs. Microservices Architectures, Client-Server Architecture, Three-Tier Architecture

Module 2: Scalability and Performance

Load Balancing Techniques: Load Balancing Algorithms (Round Robin, Least Connections, Least Response Time), Load Balancers (Hardware vs. Software), Caching Strategies: Caching layers (Client-side, Server-side, CDN), Cache invalidation strategies (Least Recently Used, Least Frequently Used), Designing for High Throughput: Asynchronous processing (message queues, event-driven architectures), Data partitioning and sharding, Optimizing database queries,

Module 3: Distributed Systems and Microservices

Distributed Systems: Challenges of distributed systems (communication, consistency, fault tolerance), Inter-process communication (RPC, message queues), Distributed databases, Microservices Architecture: Principles of microservices (small, independent services), Designing and implementing microservices, Service discovery and inter-service communication

Module 4: Fault Tolerance and Disaster Recovery

Fault Tolerance Mechanisms: Redundancy techniques (replication, failover), Circuit breakers, Retry mechanisms, Error handling and logging, Disaster Recovery Planning: Backup and recovery strategies, Disaster recovery sites (hot, warm, cold), Business Continuity Planning

Reference Books and Suggested Readings:

1. Designing Data-Intensive Applications by Martin Kleppmann

2. Building Microservices by Sam Newman
3. Release It!: Design and Deploy Production-Ready Software by Michael T. Nygard
4. Software Architecture Patterns by Mark Richards
5. System Design Interview by Alex Xu

Course Code		**Course Name**	Fundamental of Deep learning	**Course Category**	MC	Major	L	P	Pr	C
							0	3	0	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Understand the concept of a single neuron (perceptron), its limitations (linear separability), and the perceptron learning algorithm.	3
CLO2	Understand the concept of gradients, the gradient descent algorithm (batch, stochastic, mini-batch), and its role in minimizing the loss function.	4
CLO3	Understand the core concepts of CNNs (convolution, pooling), their applications in image recognition and computer vision, and common CNN architectures (LeNet, AlexNet, VGG).	3
CLO4	Derive and explain the backpropagation algorithm for training multi-layer neural networks	4

Module 1 : Foundations of Neural Networks

The Perceptron:concept, perceptron learning algorithm, Limitations of the perceptron (linear separability), Introduction to Artificial Neural Networks: Biological inspiration and the concept of artificial neurons, Feedforward neural networks: Layers, neurons, activation functions (sigmoid, ReLU, tanh), Representing data with neural networks

Module 2: Training Neural Networks

Gradient Descent: Gradient descent algorithm (batch, stochastic, mini-batch), Learning rate and its impact on training, Choosing an appropriate optimization algorithm, Backpropagation: Chain rule and backpropagation algorithm for training neural networks, Calculating gradients for weights and biases, Implementing backpropagation using a library like TensorFlow or PyTorch,

Module 3: Advanced Neural Network Architectures

Convolutional Neural Networks (CNNs): Convolutional layers, pooling layers, Applications of CNNs (image recognition, object detection), Understanding architectures like LeNet, AlexNet, VGG, Recurrent Neural Networks (RNNs):Handling sequential data (time series, natural language),Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs),Applications of RNNs (language modeling, machine translation)

Module 4:

Model Evaluation: Confusion matrix, accuracy, precision, recall, F1-score, AUC-ROC,Cross-validation techniques (k-fold, stratified k-fold), Overfitting and Regularization:Overfitting and

its causes,Regularization techniques (L1/L2 regularization, dropout),Hyperparameter Tuning**:** Hyperparameter tuning methods (grid search, random search),Techniques for efficient hyperparameter tuning (e.g., Bayesian optimization)

Reference Books:

1. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow by Aurélien Géron:
2. Deep Learning by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
3. Neural Networks and Deep Learning by Michael Nielsen
4. Pattern Recognition and Machine Learning by Christopher M. Bishop
5. Dive into Deep Learning by Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smo

Course Code		**Course Name**	Communication Skills (English)	**Course Category**	AEC	Ability Enhancement Course	L	P	Pr	C
							0	4	0	4
Pre-requisite			Nil	**Co-requisite**		Nil				

COURSE LEARNING OUTCOMES

CLO 1	Use basic English to communicate effectively in everyday situations
CLO 2	Exchange information and give instructions clearly and effectively
CLO 3	Describe past experiences, current activities, and future plans

Module 1: Everyday English Basics

- **Getting Started**- Alphabet, Colors, Nice to Meet You
- **About You**- Getting to Know You, Where Are You From?, This is My Family
- **School**- In the Classroom, At School
- **Time**- My Day, My Week, My Month
- **Shopping**- How Much Is It?, Shopping for Clothes
- **Food**- At the Supermarket, At a Restaurant, Food I Like
- **Health**- Making an Appointment, At the Doctor
- **Community**- Finding an Apartment, Around Town
- **Work**- Jobs, Getting a Job, Getting to Work, Calling in Sick

Module 2: Everyday English – 1

- **Greetings and Introductions**- Asking and Giving Personal Details, Talking About Nationality and Where You're From, Thanking and Introducing People
- **Exchanging Information**- Arranging Meetings, Talking About Work and Computers, Giving Instructions
- **Family and Friends**- Describing and Complimenting People, Talking About Art and Family
- **How Much? How Many?**- Asking Prices, Buying Food and Tickets, Talking About Food, Pets, and Quantities,
- **Describing Your Home**- Describing Apartments and Rooms, Buying Household Items, Making Plans and Suggestions
- **Describing Routines**- Daily Routines and Responsibilities, Talking About Schedules and Work
- **Things We Can/Can't Do**- Asking About Events and Dates, Expressing Feelings and Making Invitations, Shopping and Problem-Solving
- **What's Happening?**- Current Actions and Events, Talking About Activities and Hobbies

Module 3: Everyday English – 2

- **Greetings and Introductions**- Describing Hotel Rooms and Feelings, Talking About Likes, Dislikes, and Professions
- **Routines and Actions**- Talking About Tourist Attractions, Describing Locations, Directions, and Actions
- **Talking About the Past**- Past Experiences and Events, Describing Burglaries and Reporting Them

- **Past Experiences**- Talking About Hobbies and Disappointments, Keeping in Touch
- **Talking About the Future**- Weather Forecasting, Future Plans and Trips
- **Let's Trade Apartments**- Asking Favors and Buying Gifts
- Must/Mustn't and Should/Shouldn't Dos
- **Things We Have Done**- Dreams, Interests, and Life Experiences, Solving Travel Problems
- **Comparing People and Things**- Shopping, Apologizing, and Packing, Saying Goodbye and Talking About Tours

Suggested Readings

1. "English Communication for Technical Students" by M. Ashraf Rizvi
2. "English for Everyone: English Vocabulary Builder" by DK
3. "The Elements of Style" by William Strunk Jr. & E.B. White
4. "Improve Your English: English in the Workplace" by D. H. Palmer
5. "How to Speak and Write Correctly" by Joseph Devlin

Course Code		**Course Name**	Professional Skills (Career Skills)	**Course Category**	SEC	Skill Enhancement Course	L	P	Pr	C
							0	3	0	3
Pre-requisite			Nil	**Co-requisite**		Nil				

COURSE LEARNING OUTCOMES

CLO 1	Prepare a professional fit to purpose résumé in line with the job description and digital and AI-era practices
CLO 2	Prepare for job interviews
CLO 3	Participate in recruitment-related group discussions
CLO 4	Prepare self for achieving career goals through career planning and life-long learning
CLO 5	Identify career opportunities in consideration of personal potential and aspirations.

Module 1: Résumé Skills – 20 hrs

- **Résumé Skills: Preparation and Presentation** - Comprehend the importance of a résumé, identify essential components of a good résumé while preparing it.
- **Résumé Skills: Common Errors** - Identify common errors in a résumé.
- **Keywords Specific Resume** - Align resume to new-age AI-powered hiring practices
- **Skills vs Job Description** - Prepare a resume to map the job description
- **Make Specialized Resumes for Different Job Applications** - Create Resumes using AI Tools
- **Self-Presentation Even Before Interview** - Present a Video Resume
- **Work Portfolio** - Prepare a work portfolio
- **Digital Media Profiles** - Manage professional presence on digital media platforms

Module 2: Interview Skills – 15 hrs

- **Introduction to Interviews** - Describe the meaning and types of interviews.
- **Common questions** - Describe the important questions generally asked in a job interview.
- **Exchange of views**
- **Interview Skills: Preparation and Presentation** - List key interviewee skills
- **Interview Procedure** - Describe the interview procedure
- **Interview Skills: Common Errors** - Identify common errors people make during an interview.

Module 3: Interview Simulation – 8 hrs

- **Job Simulation Formats** - Critique the performance of a few simulated interviews
- **Comment Critically on Simulated Interviews** - Critique the performance of a few simulated interviews
- **Demonstrate an Ideal Interview** - Critique the performance of a few simulated interviews

Module 4: Group Discussion Skills – 10 hrs

- **Meaning and Importance of Group Discussion** - Describe the meaning and importance of a Group Discussion in a selection process.

- **Procedure of a Group Discussion** - Describe the procedure of a Group Discussion, identify essential skills to be evaluated during a Group Discussion.
- **Group Discussion: Common Errors** - Identify common errors people commit in a Group Discussion.
- **Group Discussion: Simulation** - Identify common errors people commit in a Group Discussion.

Module 5: Career Planning – 13 hrs

- **What is Career? Why a Specific Career?** - Explain the process of career development and its importance for professionals
- **Importance of Career Development** - Explain the process of career development and its importance for professionals
- **Knowing Yourself** — Personal Characteristics (MBTI - personality Test) - Explain the process of career development and its importance for professionals
- **Career Aptitude Tests** - Explain the process of career development and its importance for professionals
- **Career opportunities in Industry & Goals** - Explain the process of career development and its importance for professionals

Module 6: Exploring Career Opportunities – 10 hrs

- **Knowledge about the World of Work** - Identify career opportunities in selected fields of work
- **Sources of Career Information** - Identify career opportunities in selected fields of work
- **Skills & Career - Current Trends** - Identify career opportunities in selected fields of work
- **Process of Career Exploration** - Identify career opportunities in selected fields of work

Module 7: Lifelong Learning – 10 hrs

- **Developing Eligibility** - Develop skills and abilities to support career goals using life-long learning
- **Concept of Life-Long Learning** - Develop skills and abilities to support career goals using life-long learning
- **Sources of Life-long learning** - Develop skills and abilities to support career goals using life-long learning
- **Case Study** - Use the necessary components to prepare for a career in an identified occupation (as a case study).

SUGGESTED READINGS

1) A Course on Career Skills https://www.tcsion.com/hub/ncs/
2) Essential Career Skills https://alison.com/course/essential-career-soft-skills#google_vignette
3) Finding A Job In A Challenging Economic Times
4) https://www.communityskilling.org/pathway/finding-a-job-in-a-challenging-economic-times
5) Career Handbook https://cdn.uconnectlabs.com/wp-content/uploads/sites/123/2021/07/Career-Handbook-2019.pdf

6) The Careers Handbook: The Graphic Guide to Planning Your Future

Semester VII

Course Code		**Course Name**	Natural Language Processing (NLP)	**Course Category**	MC	Major	L	P	Pr	C
							o	3	o	3
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Explain the core tasks in NLP: Text classification, sentiment analysis, machine translation, named entity recognition, etc.	3
CLO2	Understand the importance of text preprocessing techniques (tokenization, stemming, lemmatization, stop word removal).	4
CLO3	Understand different word representation techniques (one-hot encoding, word embeddings - Word2Vec, GloVe, FastText).	3
CLO4	Utilize popular NLP libraries like NLTK, spaCy, and Transformers to perform common NLP tasks (e.g., text classification, sentiment analysis, named entity recognition).	4

Module 1: Introduction to NLP and Text Processing

What is NLP: Definition and applications of NLP, Challenges in NLP (ambiguity, variation, sparsity), Text Processing Fundamentals: Text cleaning (lowercasing, punctuation removal, stop word removal), Tokenization (word-level, character-level), Stemming and lemmatization,

Module 2: Word Representation and Language Modeling

Word Embeddings: One-hot encoding, Word2Vec (skip-gram, CBOW), GloVe, FastText, N-grams and Language Modeling: N-grams and their applications (language modeling, text prediction), Calculating probabilities using n-grams, Limitations of n-gram models,

Module 3: Syntax and Semantics

Syntax:Parts-of-speech (POS) tagging, Parsing (constituency parsing, dependency parsing), Syntactic analysis tools (NLTK, spaCy), Semantics:Word sense disambiguation,Sentiment analysis, Named Entity Recognition (NER)

Module 4: Deep Learning for NLP -

Recurrent Neural Networks (RNNs), Introduction to RNNs, LSTMs (Long Short-Term Memory), GRUs (Gated Recurrent Units), Applications of RNNs in NLP (sentiment analysis, machine translation), Transformers: Attention mechanisms, Transformer architecture (encoder-decoder), Applications of Transformers (BERT, GPT)

Module 5: NLP Applications and Ethics

NLP Applications:Text classification (sentiment analysis, spam detection), Machine translation, Chatbots and conversational AI, Information extraction, NLP Libraries: NLTK, spaCy,

Transformers (Hugging Face), Working with NLP libraries for common tasks, Ethics in NLP: Bias in NLP models, Fairness, accountability, and transparency, Ethical considerations in NLP applications

Reference Books:

1. Speech and Language Processing by Daniel Jurafsky and James H. Martin
2. Natural Language Processing with Python by Steven Bird, Ewan Klein, and Edward Loper
3. Deep Learning by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
4. "Attention Is All You Need" by Ashish Vaswani et al.
5. "Word2Vec: A Neural Language Model" by Tomas Mikolov et al
6. "Understanding Convolutional Neural Networks for NLP" by Yoon Kim.

Course Code		**Course Name**	Advance Deep Learning	**Course Category**	MC	Major	L	P	Pr	C
							2	2	0	4
Pre-Requisite			Fundamentals of Deep Learning	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy
CLO1	Recognize and analyze data with sequential dependencies (e.g., time series, natural language).	3
CLO2	Understand why traditional methods may not effectively capture temporal dependencies in sequential data.	4
CLO3	Understand the core concepts of RNNs, including the recurrent unit and the flow of information through time.	3
CLO4	Understand the concept of attention and its role in improving the performance of sequence-to-sequence models.	4

Module 1: Foundation of Sequence Data

Introduction to Sequence Data: Characteristics of sequential data (time series, natural language), Limitations of traditional machine learning models for sequential data

Module 2: Recurrent Neural Networks (RNNs)

Basic RNNs:Core concept of recurrent units and the flow of information through time, Vanishing/exploding gradients problem, Simple RNN implementations and limitations,
Advanced RNN Architectures: Long Short-Term Memory (LSTM) networks: Understanding LSTM cells (forget gate, input gate, output gate),
Gated Recurrent Units (GRUs): Architecture and comparison with LSTMs,Applications of RNNs in NLP (sentiment analysis, part-of-speech tagging)

Module 3: Sequence-to-Sequence Models and Attention Mechanisms

Sequence-to-Sequence Models: Encoder-Decoder architecture, Applications of seq2seq models (machine translation, text summarization, chatbots), Implementing basic seq2seq models,
Attention Mechanisms**:** Introduction to attention mechanisms, Types of attention (self-attention, cross-attention), Implementing attention in sequence-to-sequence models

Module 4: Transformers and Large Language Models

Transformer Architecture**:** Encoder-decoder architecture in Transformers, Self-attention mechanism in Transformers, Positional encoding,
Advanced Transformer Models**:** BERT, GPT, and other state-of-the-art transformer models,

Applications of large language models (LLMs) (text generation, conversation, code generation), Fine-tuning

Pre-trained Models**:** Fine-tuning pre-trained language models (e.g., BERT, GPT) for specific tasks

Module 5: Unsupervised Learning, Ethics, and the Future

- Unsupervised Learning in NLP: Autoencoders for text representation (e.g., variational autoencoders), Generative Adversarial Networks (GANs) for text generation, Large Language Models (LLMs): Characteristics and capabilities of LLMs, Societal impact and implications of LLMs
- **Ethics in** NLP: Bias in NLP models and datasets, Fairness, accountability, and transparency in NLP systems, Ethical considerations in the development and deployment of NLP applications,
- Future Directions in NLP: Multi-modal NLP (combining text with images, audio, etc.), Explainable AI for NLP, The future of human-computer interaction through NLP

Reference Books:

1. "Speech and Language Processing" by Daniel Jurafsky and James H. Martin
2. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
3. "Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow" by Aurélien Géron
4. "Natural Language Processing with Python" by Steven Bird, Ewan Klein, and Edward Loper
5. "Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig

Course Code		**Course Name**	Business Communication	**Course Category**	AEC	Ability Enhancement Course	L	P	Pr	C
							4	0	0	4
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Discuss the scope and complexity of business communications	
CLO2	Carry out effective interpersonal communication, including the use of listening skills, verbal skills and non-verbal communication	
CLO3	Apply practical techniques for effective group communication, including presentations, interviews, meetings, conferences, public relations	
CLO4	Carry out written business correspondence including letters, documents reports, etc.	
CLO5	Communicate effectively with the help of digital media including emails, virtual meetings and social media posts	

Module 1: Overview of Business Communications

Introduction to Business communication - Introduction, Scope of Business Communications, Complexity of Business Communications, Importance of Communication for Business, Impact of Poor Communication, Definition and Introduction to Business Communication, Types & Levels of Business Communication, Types of Business Communication, Levels of Business Communication, Real-Life Examples-Based Exercises for Practice.

Dimensions of communication in an Organisation - Introduction, Discussing the Scope and Complexity of Business Communications, Common Dimensions of Communication in an Organization, Common Channels of Communication in an Organization, Real-Life Examples-Based Exercises for Practice.

Channels of communication in an organization - Introduction, Discussing the Scope and Complexity of Business Communications, Understanding and Applying the Level of Communication in Vertical and Horizontal Hierarchy.

Barriers to Business Communication - Introduction, Discussing the Scope and Complexity of Business Communications, Discussing the Barriers Observed in Effective Business Communications, Engaging in Real-Life Examples-Based Exercises to Overcome Communication Barriers.

Module 2: Interpersonal Skills

Effective interpersonal communication - Introduction, Carrying Out Effective Interpersonal Communication, Describing Effective Interpersonal Communication, Applying Knowledge through Real-Life Examples-Based Exercises.

Listening Skills - Introduction, Carrying Out Effective Interpersonal Communication, Applying Effective Listening Skills, Enhancing Practical Mastery through Real-Life Examples-Based Exercises.

Speaking Skills - Introduction, Carrying Out Effective Interpersonal Communication, Applying Effective Speaking Skills, Reinforcing Practical Mastery through Real-Life Examples-Based Exercises.

Loud Reading Skills - Introduction, Understanding Non-Verbal Cues, Impact of Non-Verbal Communication, Developing Effective Non-Verbal Communication Skills, Real-Life Examples and Exercises.

Module 3: Body Language & Kinesics

Non-Verbal Communication - Introduction, Reading Beyond the Surface, Reading & Interpersonal Communication, Decoding the Unspoken, Real-Life Examples and Exercises.

Module 4: Group Communications

Principles of group communication - Introduction, Core Principles, Applicability Across Settings, Real-Life Exercises.

Effective presentations - Introduction, Building a Winning Presentation, Time Management: Your Key Ally, Real-Life Examples and Exercises.

Effective Meetings and conferences - Introduction, Building Effective Gatherings, Real-Life Examples and Exercises.

Effective Interviews - Introduction, Preparation: Fueling Your Confidence, Shining in the Spotlight: Commanding the Conversation, Adapting to Diverse Stages: Navigating Different Dynamics, Beyond the Conversation: Leaving a Lasting Impression, Real-Life Exercises for Success.

Module 5: Written Business Correspondence

Letter Writing - Introduction, Core Principles for Clarity and Impact, Exploring Diverse Forms of Correspondence, Ethical Considerations and Cultural Awareness, Real-Life Exercises for Growth.

Report Writing - Introduction, Key Principles, Types of Business Reports, Real-Life Exercises.

Documentation maintenance - Introduction, The Value of Good Maintenance, Navigating the Physical and Digital, Mastering the Maze: Organization and Categorization, Version Control: Preserving the Chain of History, Safeguarding the Knowledge: Security Measures, Real-Life Exercises.

Module 6: Digital Business Correspondence

Email Etiquette - Introduction, The Pillars of Professionalism, To, CC, and BCC, Subject Line, Opening Salutation, Closing Salutation, Pro Tip.

Virtual Meetings & Netiquette - Introduction, Choosing the Right Platform, Preparation is Key, Communication and Netiquette, Building Rapport and Collaboration, Real-Life Exercises, Visual Storytelling, Collaborative Learning, Expanding Your Toolbox, Examples for Real-Life Exercises.

Digital Work Collaboration - Notion, etc.- Introduction, Building Bridges, Not Walls, The Digital Toolkit, Mastering the Symphony, Real-Life Exercises, Engaging Activities, Expanding Your

Toolkit.

Module 7: Social Media Communications & Digital Marketing

Introduction to social media communications - Discuss the various social media communications used in business and digital tools and best practices associated with them.

Introduction to digital marketing - Discuss the various social media communications used in business and digital tools and best practices associated with them.

Reference Books and Suggested Readings

1. Guffey, Mary Ellen, and Dana Loewy. **Business Communication: Process and Product**. Cengage Learning.
2. Thill, John V., and Courtland L. Bovee. **Excellence in Business Communication**. Pearson Education.
3. Rentz, Kathryn, Paula Lentz, and Anupama Pradhan. **Business Communication: Making Connections in a Digital World**. McGraw Hill.
4. Bovee, Courtland L., and John V. Thill. **Business Communication Essentials**. Pearson Education.
5. Murphy, Herta A., Herbert W. Hildebrandt, and Jane P. Thomas. **Effective Business Communication**. McGraw Hill.

Semester VIII

Course Code		**Course Name**	Project Dissertation	**Course Category**	Major	Major	L	P	Pr	C
							0	12	0	12
Pre-Requisite			Nil	**Co-Requisite**		Nil				

Course Learning Outcomes:

CLO No.	At the end of the course the learners will be able to:	Bloom's taxonomy (Bt) Level
CLO1	Identify and refine a relevant research question.	1
CLO2	Apply research design principles to the question and select an appropriate methodology.	3
CLO3	Design and manage an original piece of project work.	6
CLO4	Select appropriate methodologies, methods, and forms of analysis to create a research design and justify the choices made.	5
CLO5	Discuss the ethical aspects of your research and obtain ethical approval if necessary.	4
CLO6	Synthesize prior knowledge and skills and apply them to conduct a comprehensive study.	5
CLO7	Establish connections between theoretical frameworks and methods in your area of study.	4
CLO8	Present your findings in a suitable written format.	6

Guidelines for Project Dissertation:

1. Selection of Topic

- Choose a project aligned with AI & ML applications. Examples include:
    - Predictive modeling (e.g., stock prices, disease outbreaks).
    - Natural Language Processing (NLP) (e.g., chatbots, sentiment analysis).
    - Computer Vision (e.g., image classification, object detection).
    - Reinforcement Learning (e.g., game AI, robotics).
- Ensure the topic is innovative, feasible, and relevant to industry trends or research.

2. Problem Statement

- Clearly define the problem you aim to solve.
- Highlight the importance, scope, and impact of the problem in the AI/ML domain.
- Provide context using industry challenges, academic gaps, or case studies.

3. Literature Review

- Conduct a detailed review of existing research and applications in the chosen area.

- Identify knowledge gaps and justify how your project contributes to addressing these gaps.
- Use academic journals, white papers, and reputable online resources for references.

4. Methodology

- Describe the methodology in detail:
    - Dataset Preparation: Mention data sources (e.g., Kaggle, UCI), preprocessing steps (cleaning, normalization), and augmentation techniques.
    - Algorithms and Models: Specify the ML models used (e.g., decision trees, neural networks) and justify your choice.
    - Tools and Frameworks: Use AI/ML tools like Python, TensorFlow, PyTorch, Scikit-learn, or OpenCV.
    - Evaluation Metrics: Define metrics such as accuracy, F1 score, or mean squared error to evaluate model performance.

5. Implementation

- Detail the implementation process, including:
    - Setting up the development environment.
    - Training and testing models.
    - Optimizing algorithms (e.g., hyperparameter tuning, regularization techniques).
- Include architectural diagrams, code snippets, or flowcharts to explain your implementation.

6. Results and Analysis

- Present the outcomes of your project:
    - Comparison of model performance with benchmarks.
    - Visualizations (e.g., confusion matrices, ROC curves, or loss graphs).
    - Observations and insights gained from your analysis.
- Discuss any limitations and propose ways to address them.

7. Ethical Considerations

- Address ethical aspects such as:
    - Ensuring data privacy and avoiding biases in the dataset.
    - Transparency in AI/ML decision-making.
    - Compliance with relevant standards or regulations.

8. Discussion and Conclusion

- Relate your results to the original objectives.
- Discuss how your findings contribute to the AI/ML domain.
- Propose potential future enhancements or applications of your project.

9. Report Formatting

- Follow a standard structure:
    - Title Page: Project title, student name, guide name, and institution details.

- Abstract: A brief summary of your project (150–250 words).
- Introduction: Background, problem statement, and objectives.
- Literature Review: Summary of previous work in your area.
- Methodology: Dataset, algorithms, and tools used.
- Implementation: Process description and system architecture.
- Results and Discussion: Findings and their interpretation.
- Conclusion: Summary, implications, and future work.
- References: Use a consistent citation style (e.g., APA, IEEE).

10. Tools and Resources

- Use tools like GitHub for version control and Jupyter Notebook for code development.
- Experiment with cloud platforms (e.g., AWS, Google Cloud, Azure) for scalability.
- Leverage pre-trained models (e.g., BERT, ResNet) for complex tasks.

11. Presentation

- Prepare a concise and visually engaging presentation to defend your dissertation:
  - Include problem context, methodology, results, and contributions.
  - Use visuals like graphs, charts, and flow diagrams for better comprehension.